

Appendix C to the 2015 Georgia Milestones Growth Model
Report:
Investigation of Potential Ceiling and Floor Effects.

Adam R. VanIwaarden

Damian W. Betebenner

National Center for the Improvement of Educational Assessment (NCIEA)

April 2016

1 Introduction

In the 2014-2015 academic year Georgia transitioned from its previous assessment program, the Criterion-Referenced Competency Tests (CRCT), to the [Georgia Milestones Assessment System](#). The transition included numerous changes to the assessment system including the incorporation of new performance standards. As other states have gone through similar assessment transitions, many have observed ceiling and floor effects in the new assessments (i.e. a relatively large proportion of students scoring at/near the scale extremes). This has occurred despite purported improvements in assessment qualities that should prevent these effects (e.g. adaptive tests). Regardless of the source of assessment ceilings/floors, they can telegraph onto the Student Growth Percentile (SGP) calculations making these estimates questionable.

Although very similar in nature, ceiling effects are somewhat more problematic than floor effects because students that consistently achieve the highest scores receive lower than expected growth percentiles and are therefore negatively impacted. Conversely, the consistently lowest achieving students have higher estimated SGPs than would be expected. Although this could be interpreted as a positive impact for these students by giving them higher SGPs, it can also conceal unacceptably low growth that needs to be identified and addressed.

Essentially these problems are caused by the way in which a “percentile” is most typically defined to begin with, and the inability of the assessments (and therefore the SGP model) to make granular distinctions between kids who score at the extremes of the test year after year. As an example, if a group of students were tested with a relatively easy test and 20% of the students had a perfect score, these students would be defined as being in the 80th percentile because they scored higher than 80% of their peers. This is somewhat misleading however, because their score was *equal to* or greater than *all* of their peers and so could be also described as achieving at the 99th percentile under an equally valid definition.

To extend this heuristic from achievement to growth, if 50% of those top students also scored perfectly on the next test, we might estimate that they had 50th percentile growth. Although there is nothing *technically* incorrect about this estimate since their growth is fairly typical for their academic peer group, it is an inadequate or unsatisfactory assessment of their growth because they have consistently attained at the highest observable level. Furthermore, if it is typical for their peers to maintain perfect scores then even small deviations from a perfect score could produce *low growth* SGP estimates. Ideally, the expectation is that the majority of SGPs for students scoring at or near the lowest obtainable scale score (LOSS) will be low (preferably less than 5 and not higher than 10), and that SGPs for students with the highest obtainable scale score (HOSS) will be high (higher than 95 and not less than 90).

Given these impacts and the difficulty in detecting them using traditional SGP diagnostic tools, the Center for Assessment has added “Ceiling/Test Effects” indicators to the SGP model goodness of fit plots and is providing all clients even more rigorous diagnostic and descriptive analyses through this Appendix to the annual technical report. This report includes:

1. Plots of the scale score distributions for the current and prior years’ tests, which may provide an indication of whether a ceiling or floor is present in either (or both) the current or historical data.
2. Box plots showing the range and distribution of SGPs for *only* the highest and lowest achieving students in the current year.

2 Prior and Current Year Score Distributions

The marginal and conditional distributions of scale scores can serve as a preliminary indicator of potential ceiling or floor effects in the calculation of student growth percentiles. Some minor problems could present themselves if these characteristics are present in either prior or current year scores, and are particularly likely when present in both. The plots below depict distributions for the current year and the most recent prior year used in the SGP calculations. The marginal (individual or univariate) distributions for each year are shown in the first subsections below, followed by the conditional (joint) distributions.

2.1 Marginal Distributions

Generally there is little evidence of potential issues with the Georgia tests in either the prior (2014 CRCT, shown in blue) and current (2015 Milestones, shown in red) end of grade (EOG) test scale scores. There are some indications of potential minor ceiling effects in the 2014 CRCT assessments, particularly the Reading data. These appear as spikes in the distributions at the extremes. The 2015 Milestones distributions are not concerning (although the tails of the distributions are long). This conforms to what was first observed in the SGP model goodness of fit plots (see 2015 technical report Appendix A), and conditional density and SGP distribution box plots in subsequent sections of this Appendix.

2.1.1 ELA

Fig. C.1: Marginal distributions of prior scale scores: CRCT Reading.

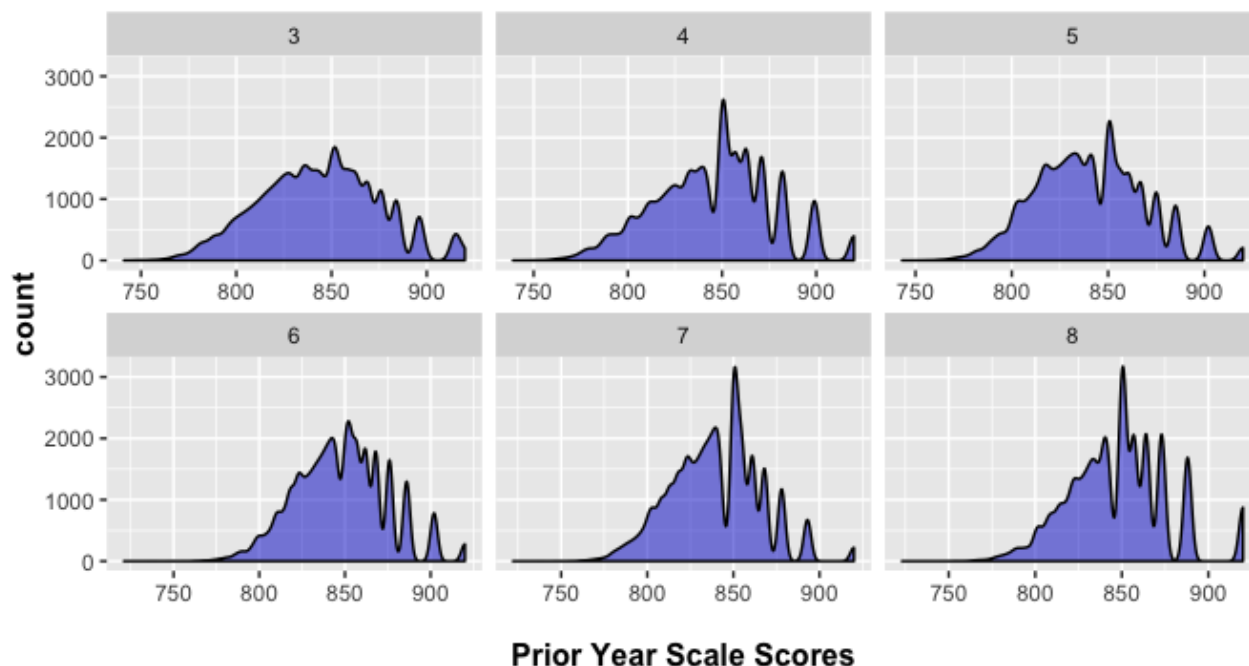
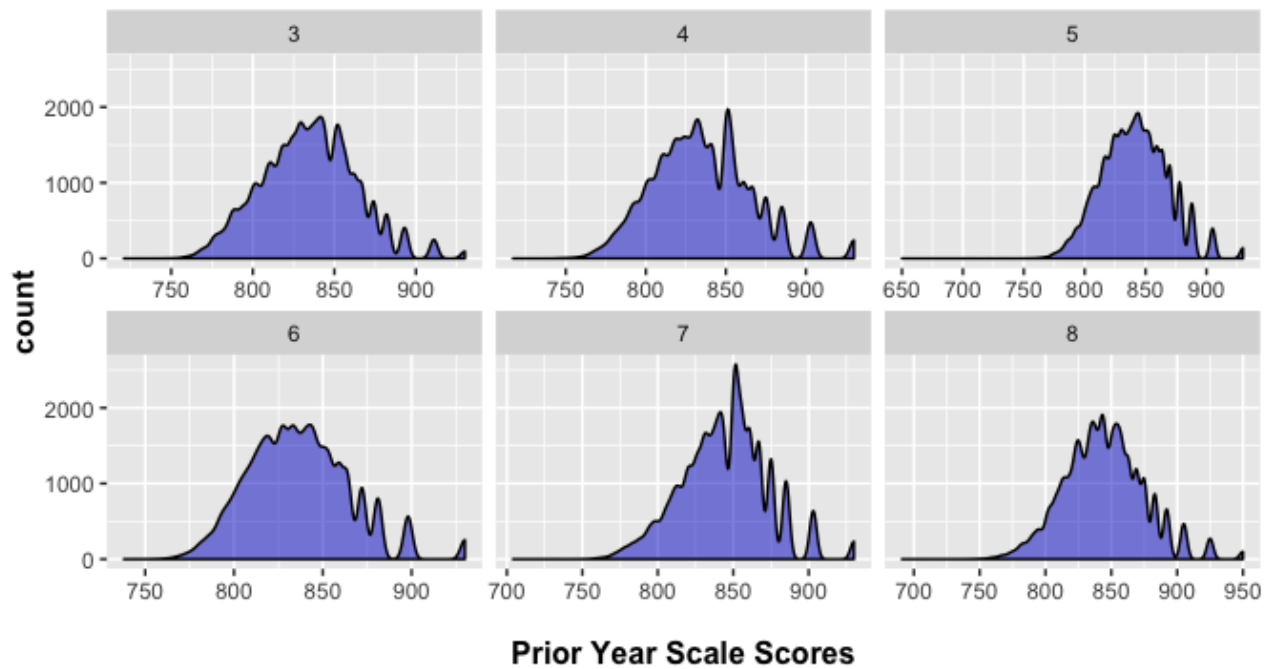
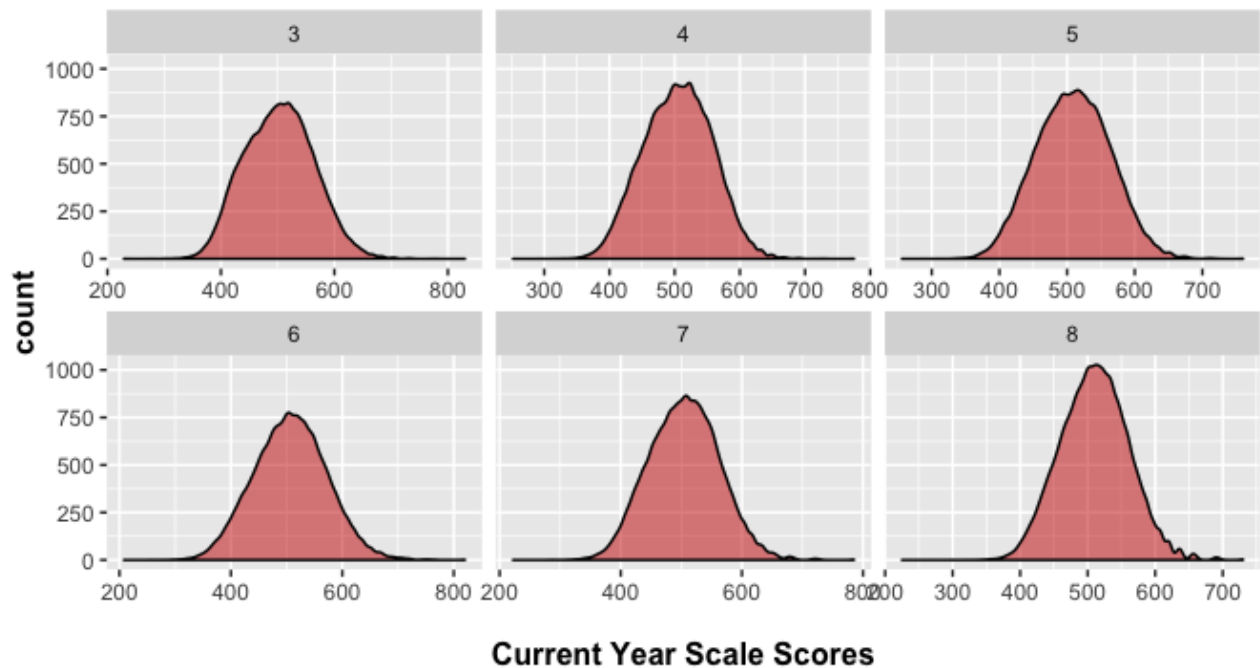


Fig. C.2: Marginal distributions of prior scale scores: CRCT ELA.**Fig. C.3:** Marginal distributions of current scale scores: EOG ELA.

2.1.2 Social Studies

Fig. C.4: Marginal distributions of prior scale scores: CRCT Social Studies.

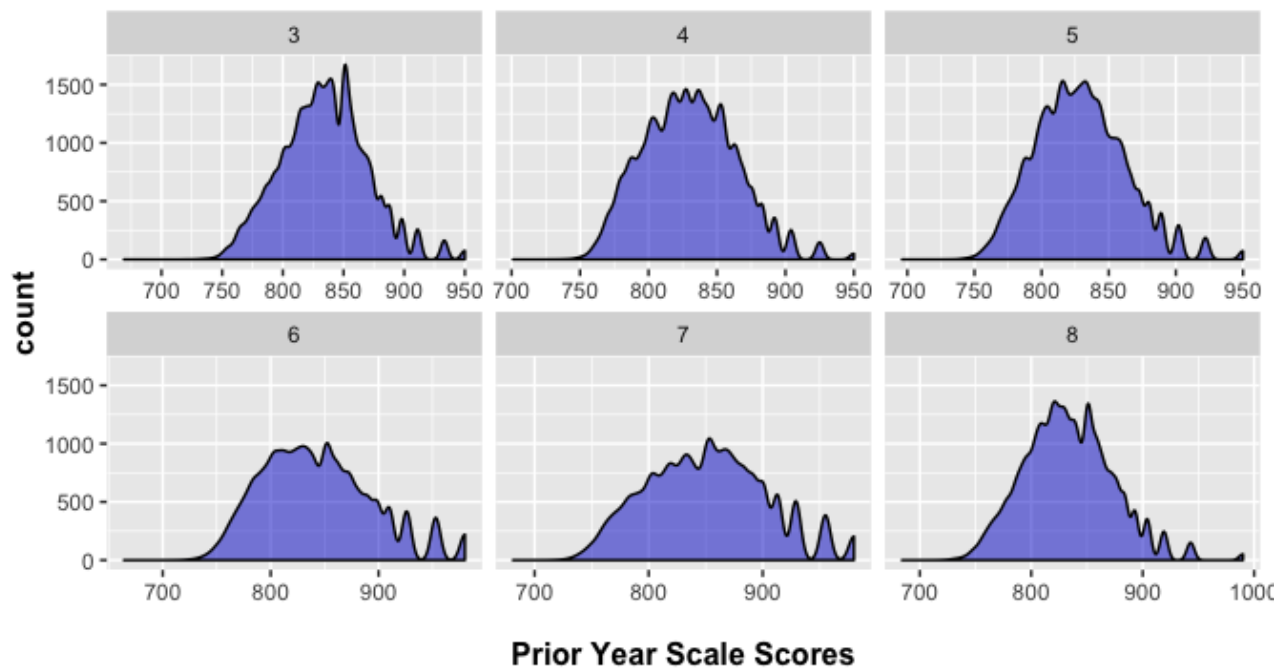
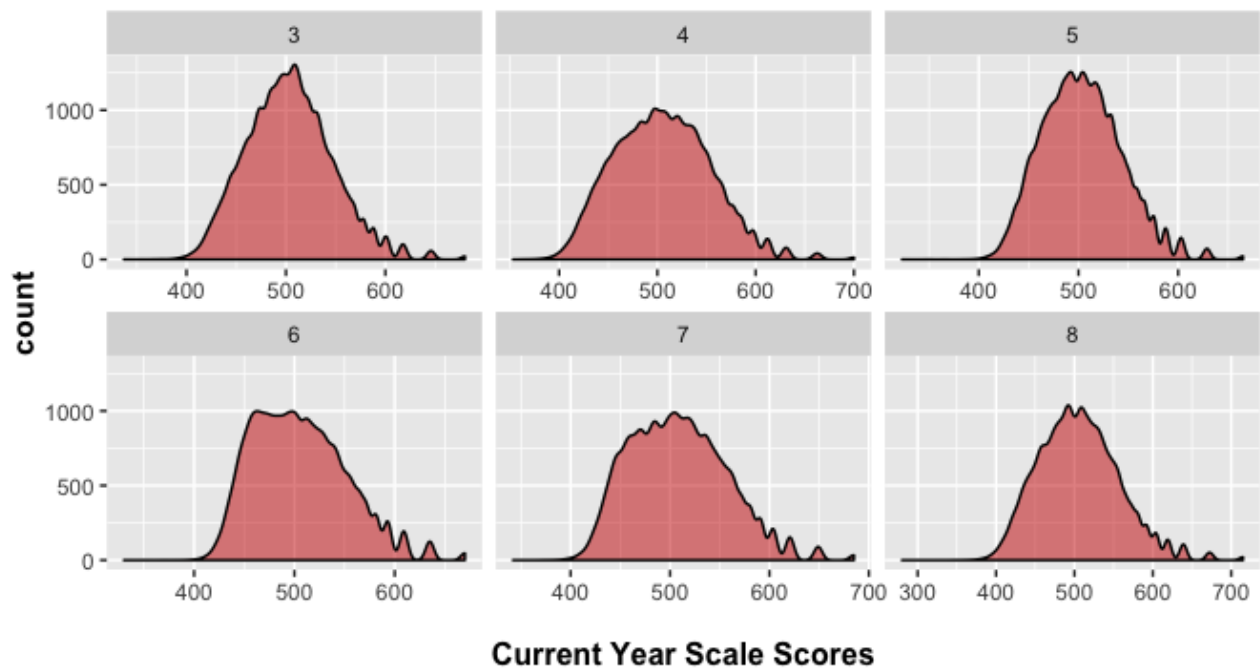
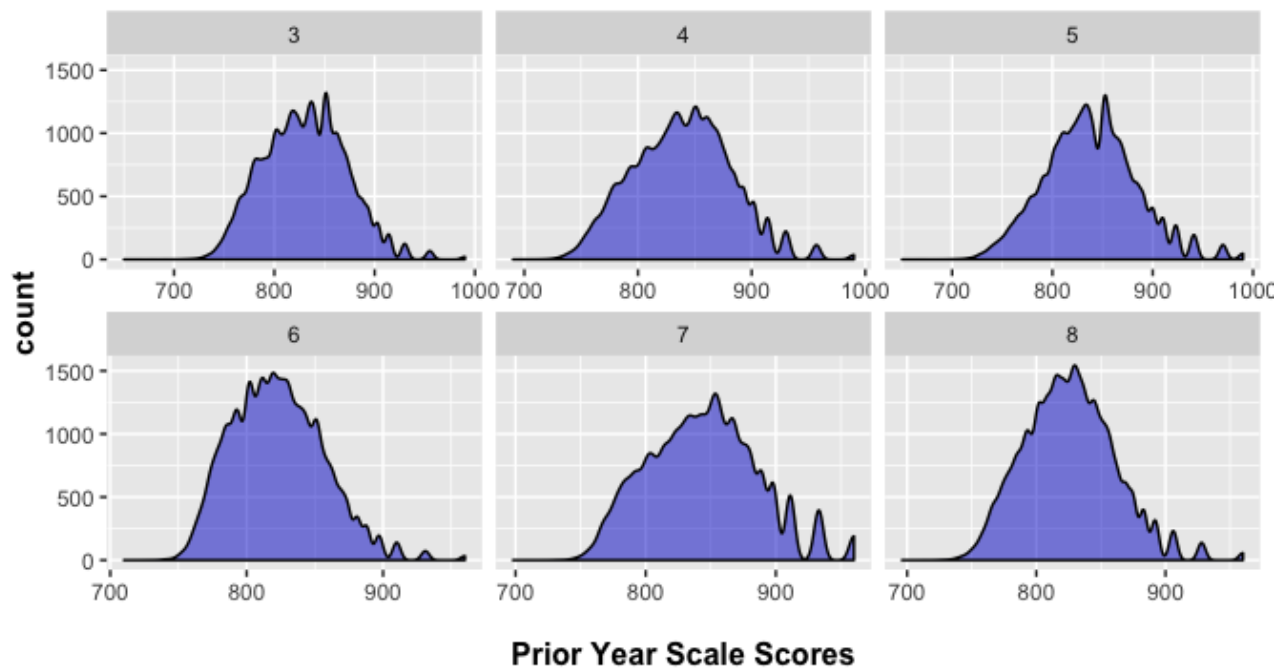
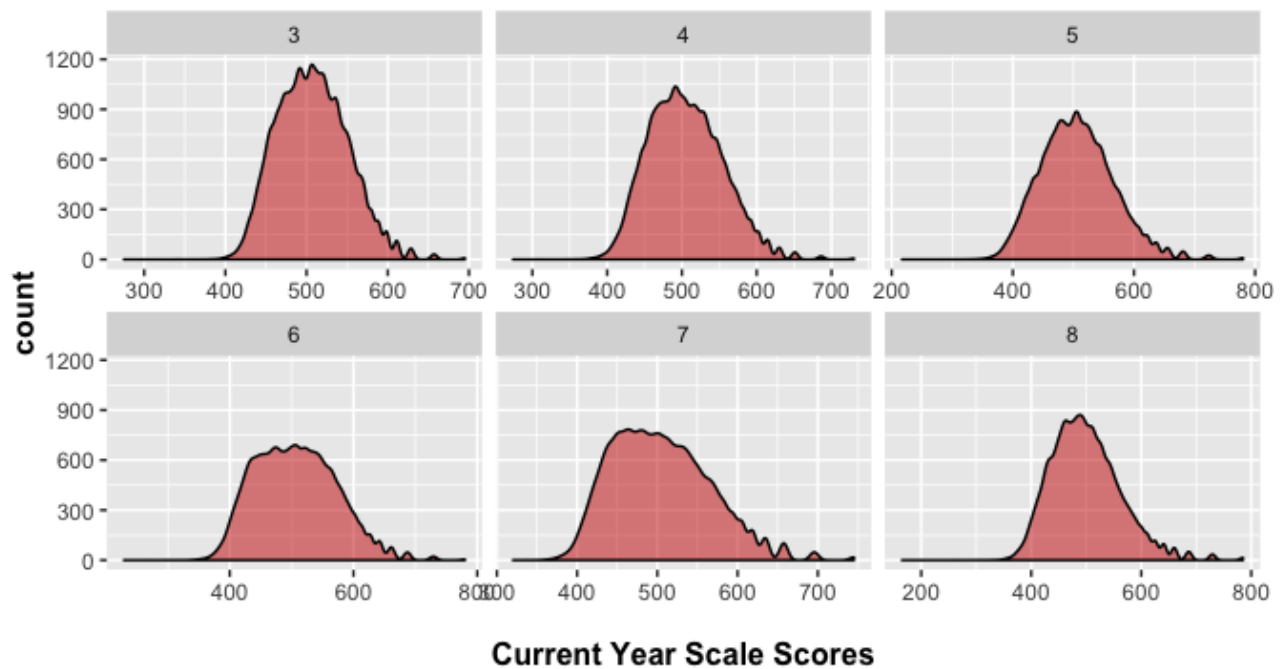


Fig. C.5: Marginal distributions of current scale scores: EOG Social Studies.



2.1.3 Science

Fig. C.6: Marginal distributions of prior scale scores: CRCT Science.**Fig. C.7:** Marginal distributions of current scale scores: EOG Science.

2.1.4 Mathematics

Fig. C.8: Marginal distributions of prior scale scores: CRCT Mathematics.

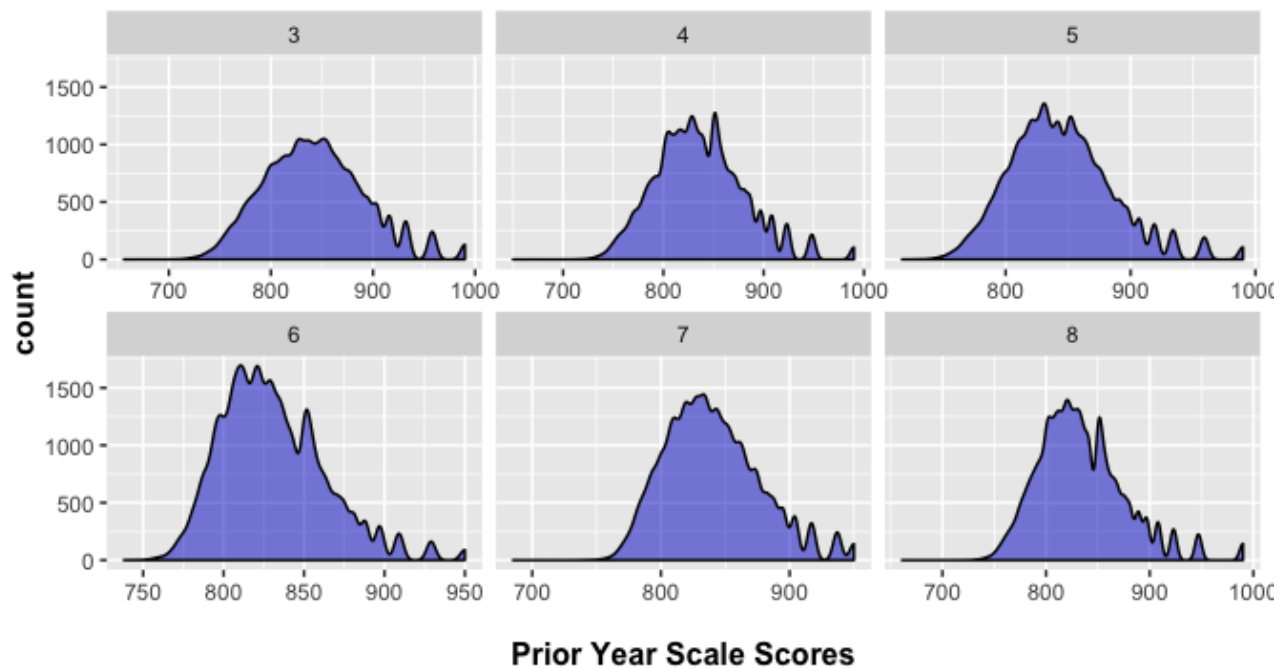
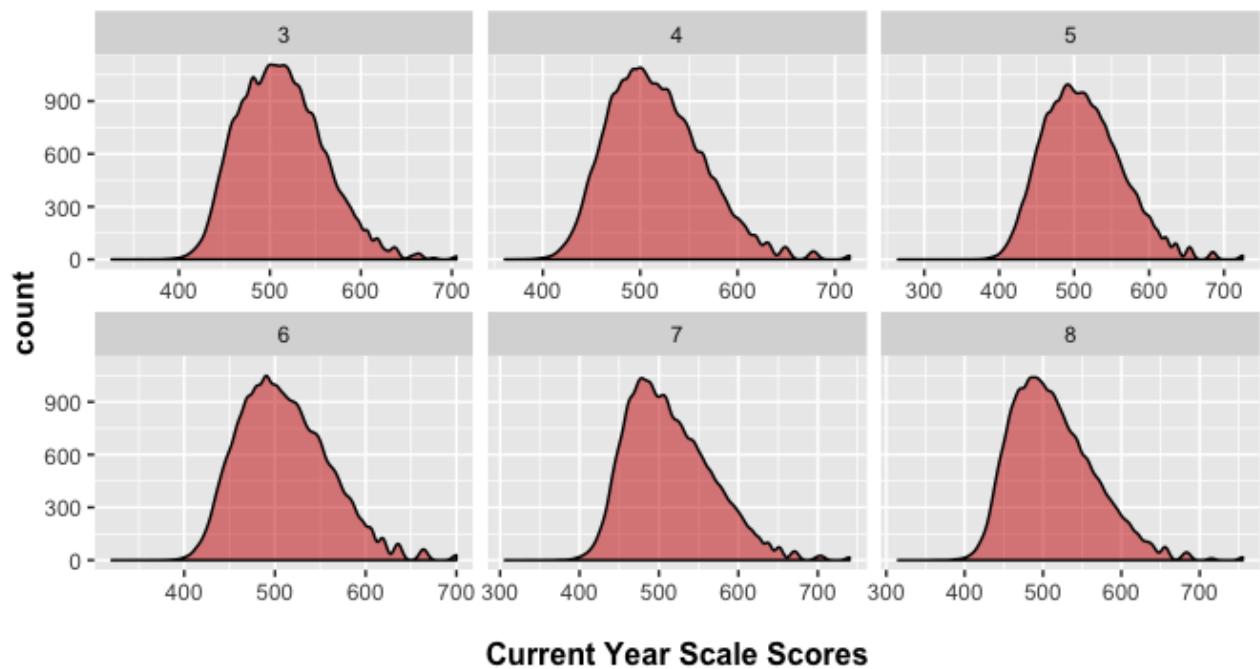


Fig. C.9: Marginal distributions of current scale scores: EOG Mathematics.



2.1.5 End-of-Course Test Subjects

Fig. C.10: Marginal distributions of prior scale scores: End-of-Course Subjects.

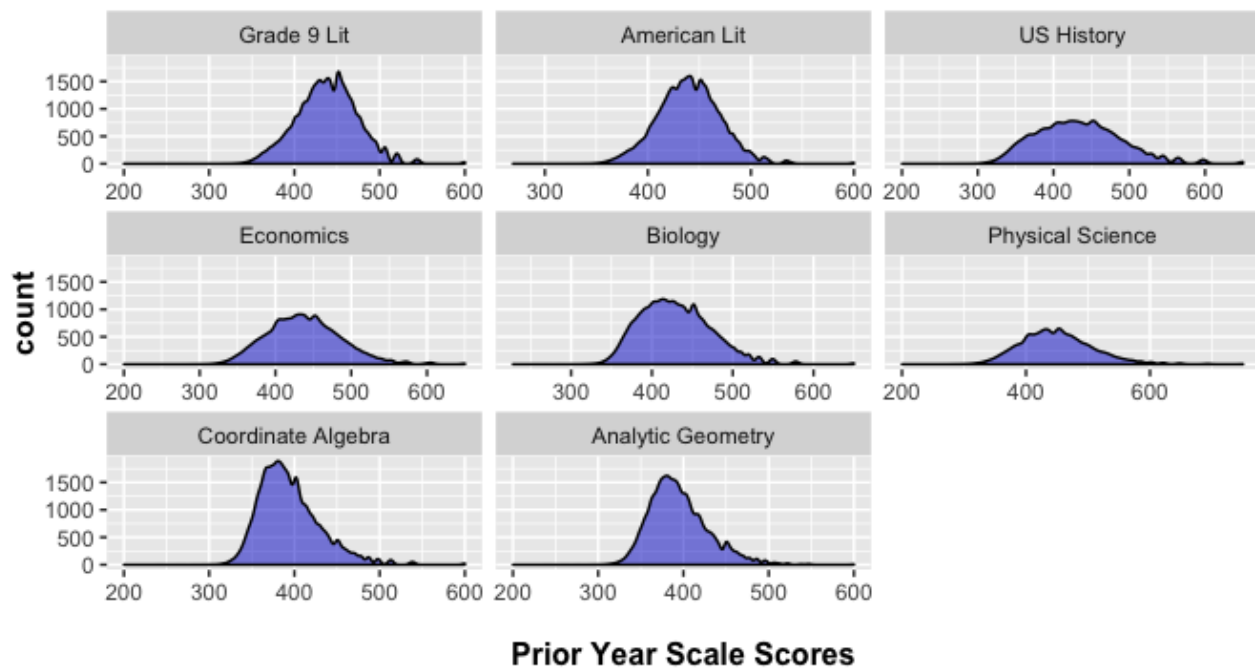
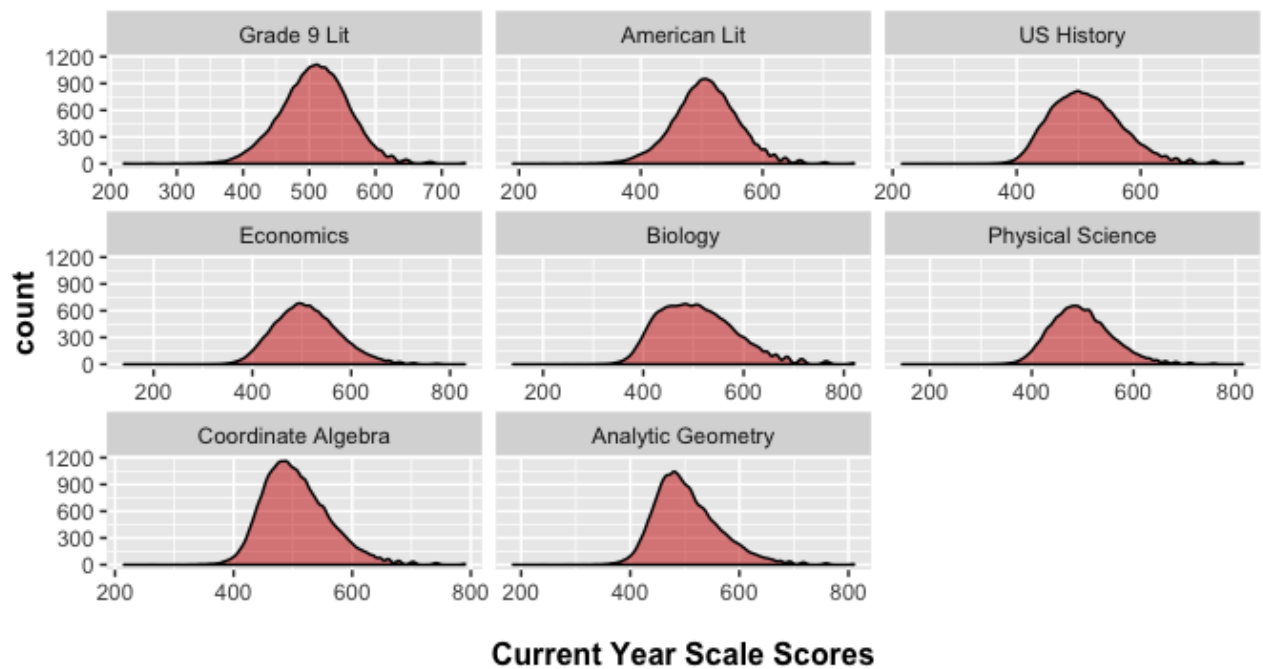


Fig. C.11: Marginal distributions of current scale scores: End-of-Course Subjects.



2.2 Conditional Distributions

The marginal density plots provide a limited amount of information, particularly for the potential for ceiling/floor effects in the calculation of *growth*. In order to provide a more nuanced view of the relationship between the prior and current scale scores, the following plots depict the conditional (joint) distributions for each content area and grade level. These plots start with a basic scatter plot of each student's scores, and on top of this is layered 1) **green contour lines** to provide a sense of the joint density, 2) a **magenta non-linear line** identifying the bivariate relationship between prior and current scores, and 3) **rug plots** that describe the marginal distributions (as above, the prior scores are blue and current scores are red).

Ceiling or floor effects can appear as dark shaded points in the extreme top-right or bottom-left corners of the plots. This suggests that staying at the extremes from year to year is not uncommon, which may lead to odd growth estimates for these chronically high/low achieving students. For the 2014 and 2015 Georgia data, we again see very few issues in all content area and grade combinations. Where minor ceiling/floor effects appear in the prior year (CRCT) data, the 2015 score distributions for these students are well distributed. That is, in plots in which a minor shaded point in the upper right corners are present the other students who scored at the HOSS on the 2014 CRCT tests had a wide range of Milestones scores, which lessens the concern for a ceiling effect in most cases.

2.2.1 End-of-Grade Content Areas

Fig. C.12: Conditional distribution(s) of current and prior scale scores: ELA.

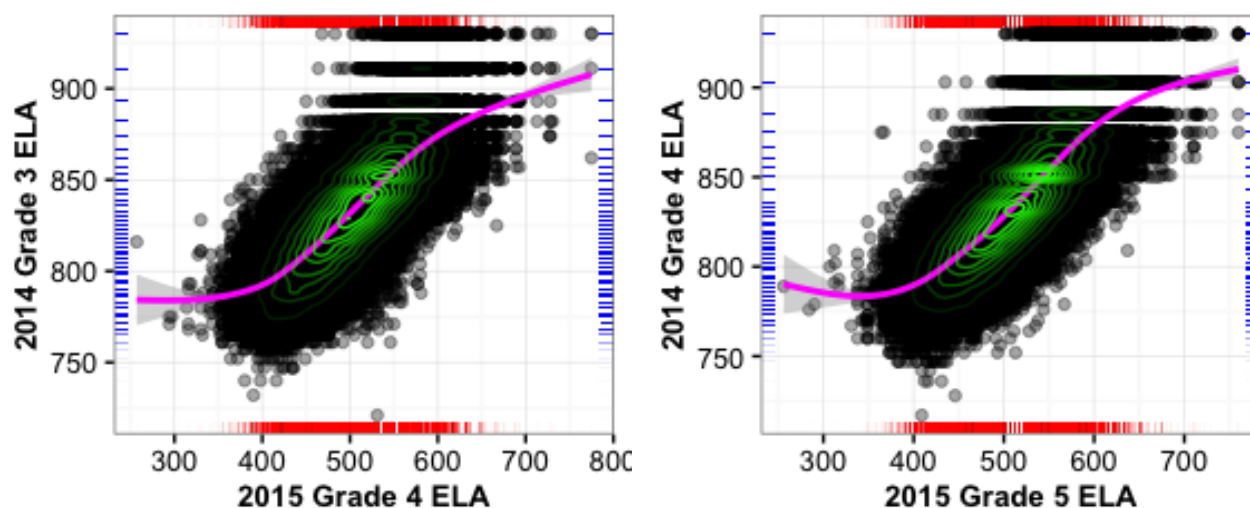


Fig. C.13: Conditional distribution(s) of current and prior scale scores: ELA *Continued*.

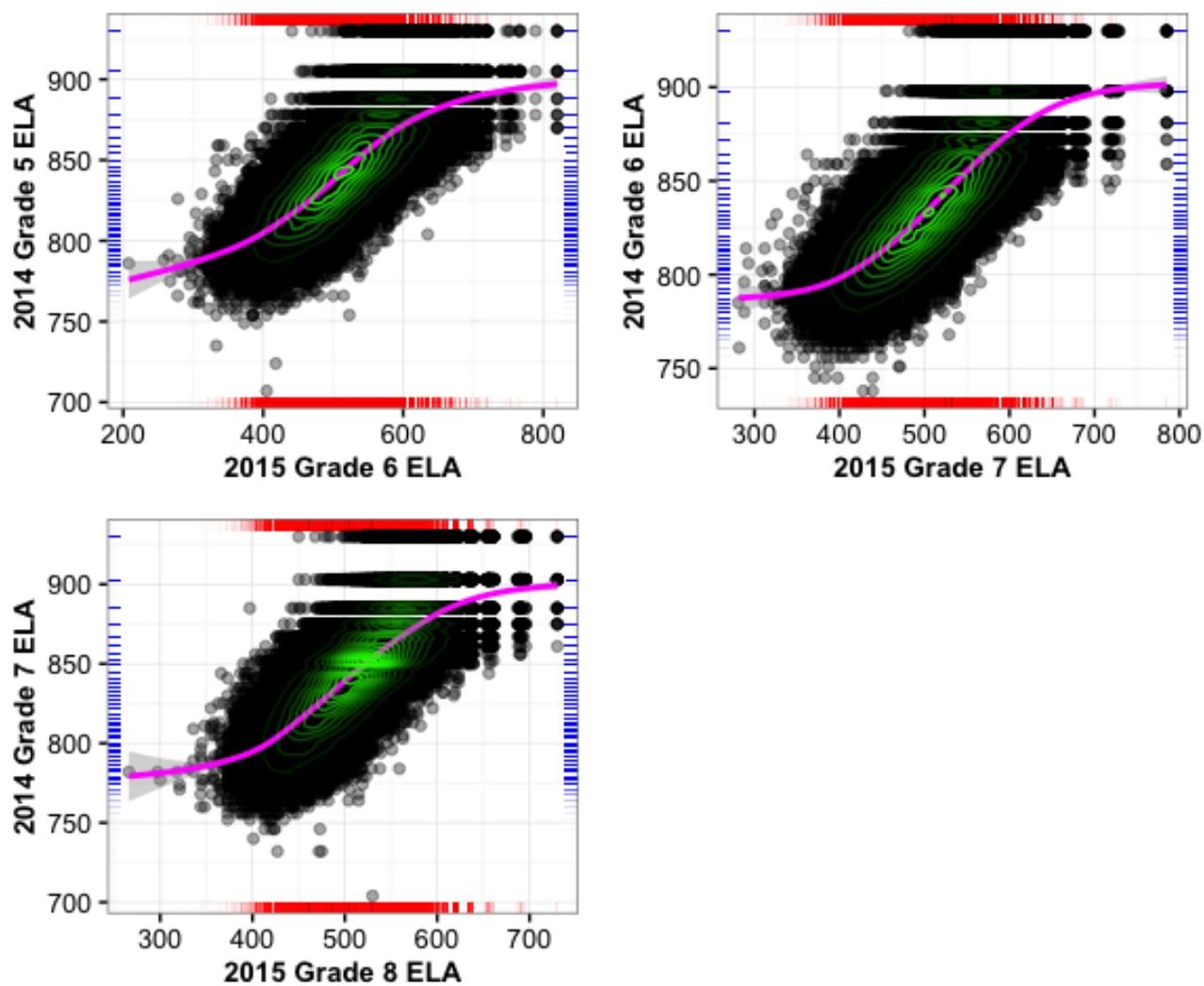


Fig. C.14: Conditional distribution(s) of current and prior scale scores: Social Studies.

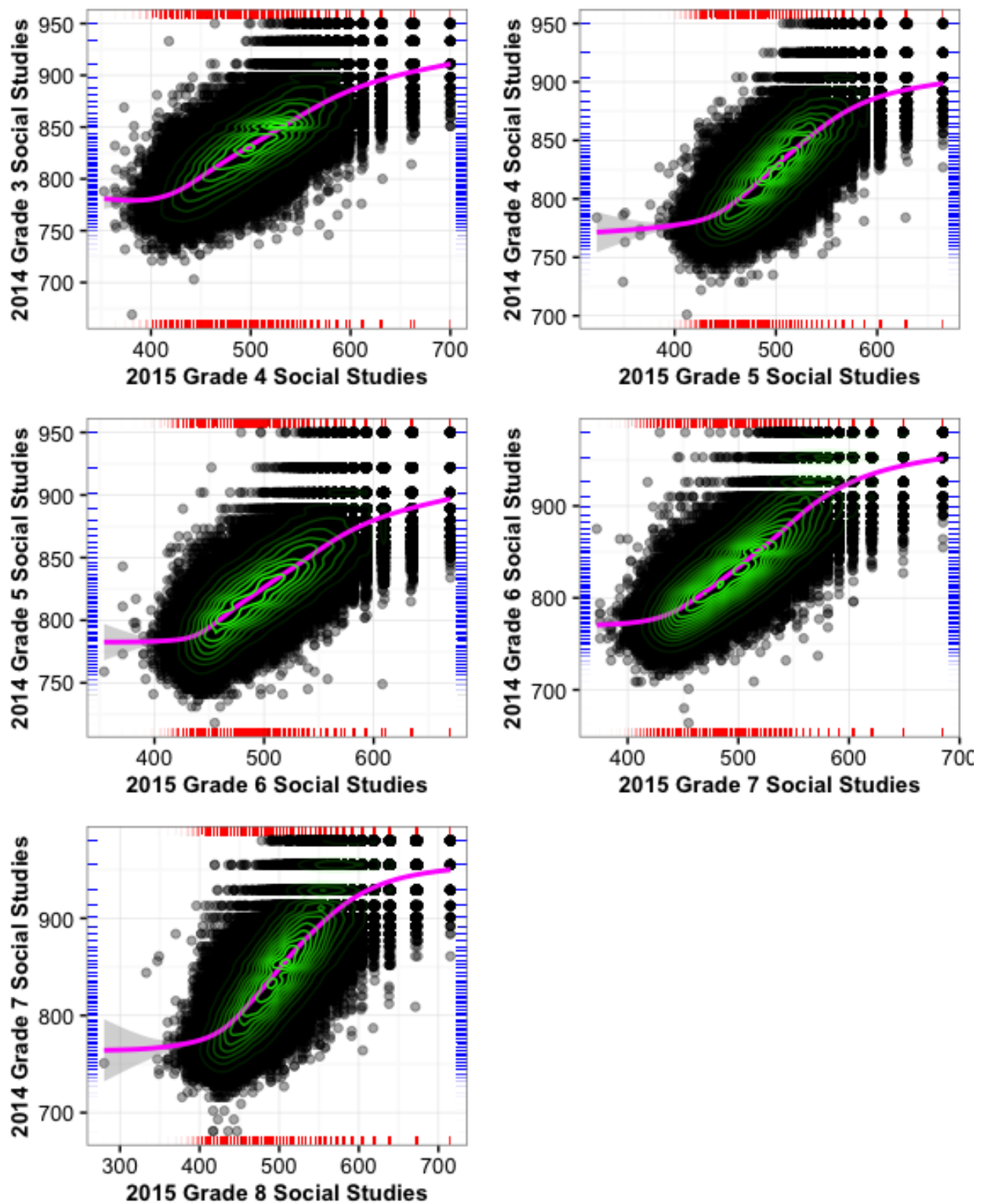


Fig. C.15: Conditional distribution(s) of current and prior scale scores: Science.

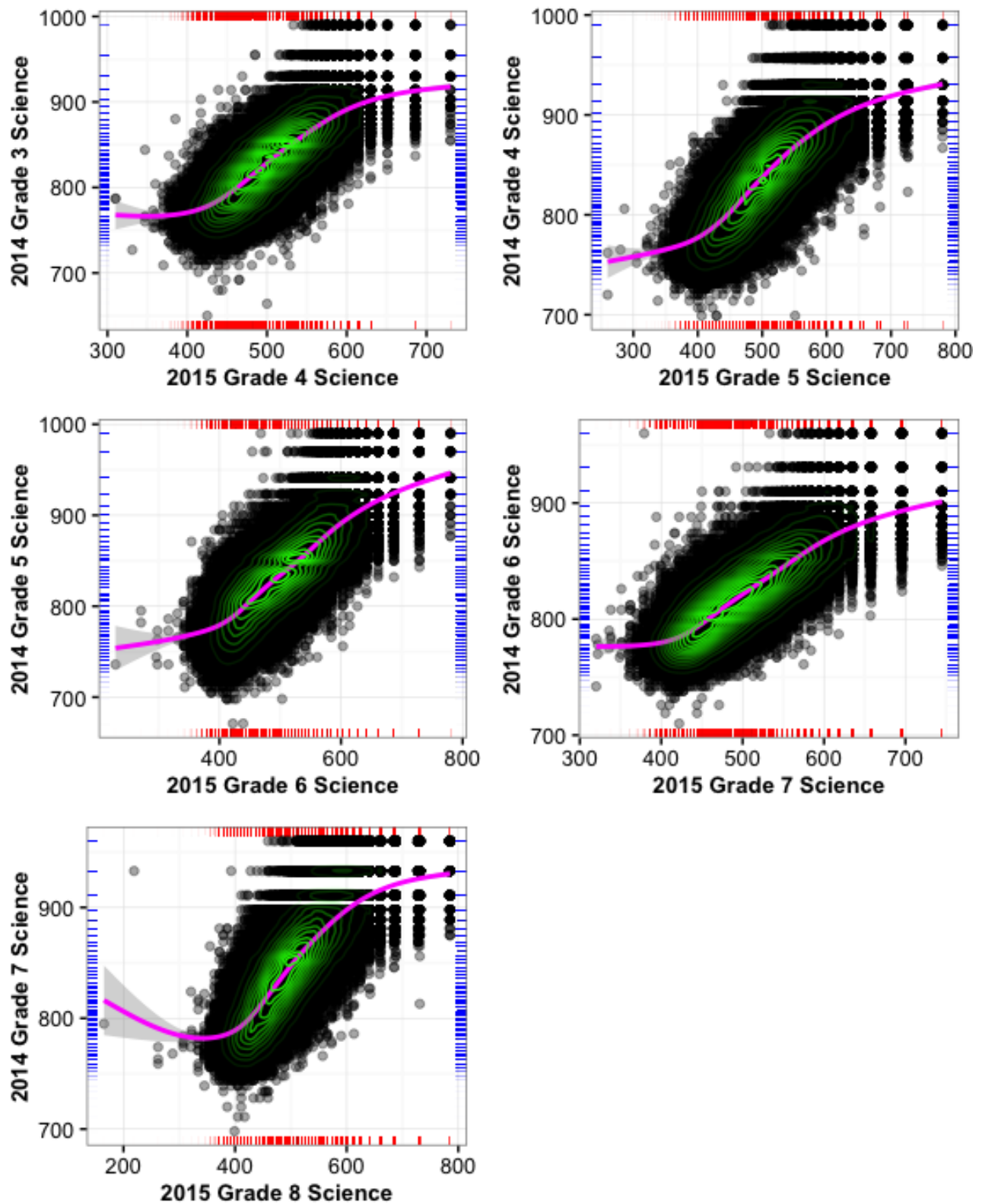
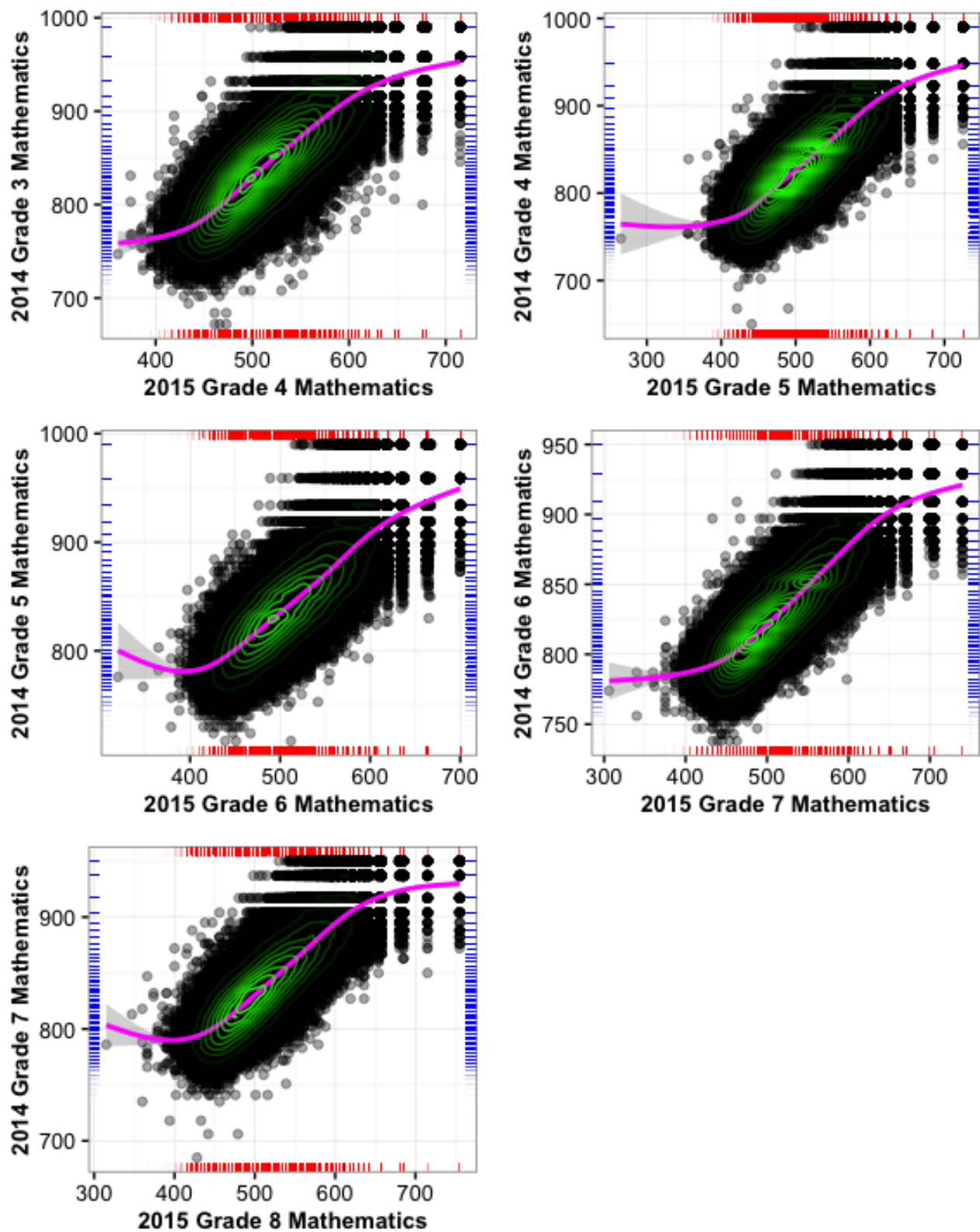


Fig. C.16: Conditional distribution(s) of current and prior scale scores: Mathematics.



2.2.2 End-of-Course Test Subjects

The conditional density plots for the EOC test subjects are displayed below. The most recent prior is used in each plot to provide insight on each academic peer (norm) group that were analyzed separately. Note the bottom-left plot in Figure C.17 below for Grade 9 Literature course repeaters, particularly the point in the upper-right hand corner of the plot. This is a potential indicator for a ceiling effect issue. Several other individual plots in other subjects also display a similar indicators, including other course repeaters and progressions with skipped years.

Fig. C.17: Conditional distribution(s) of current and prior scale scores: 9th Grade Lit.

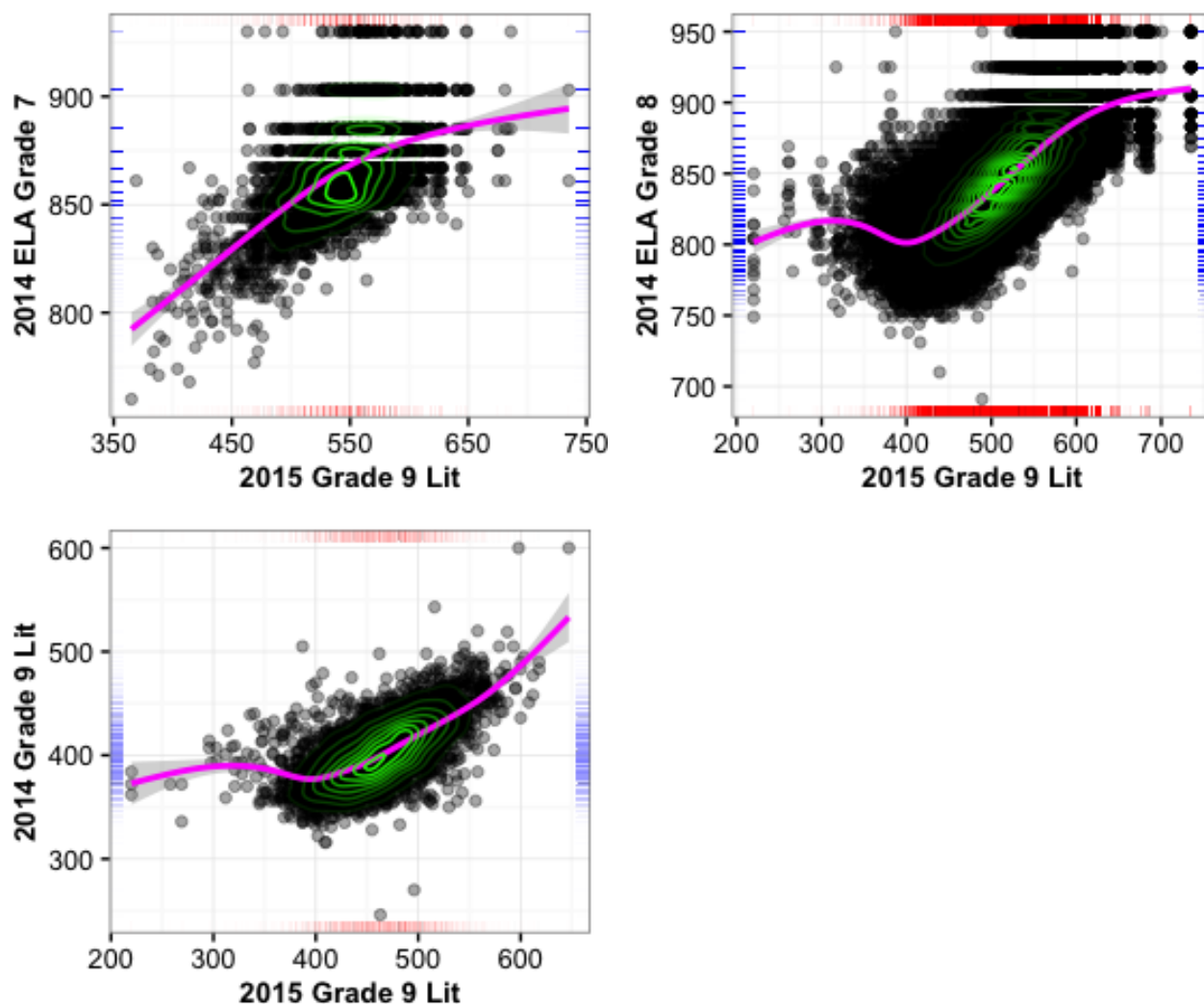


Fig. C.18: Conditional distribution(s) of current and prior scale scores: American Lit.

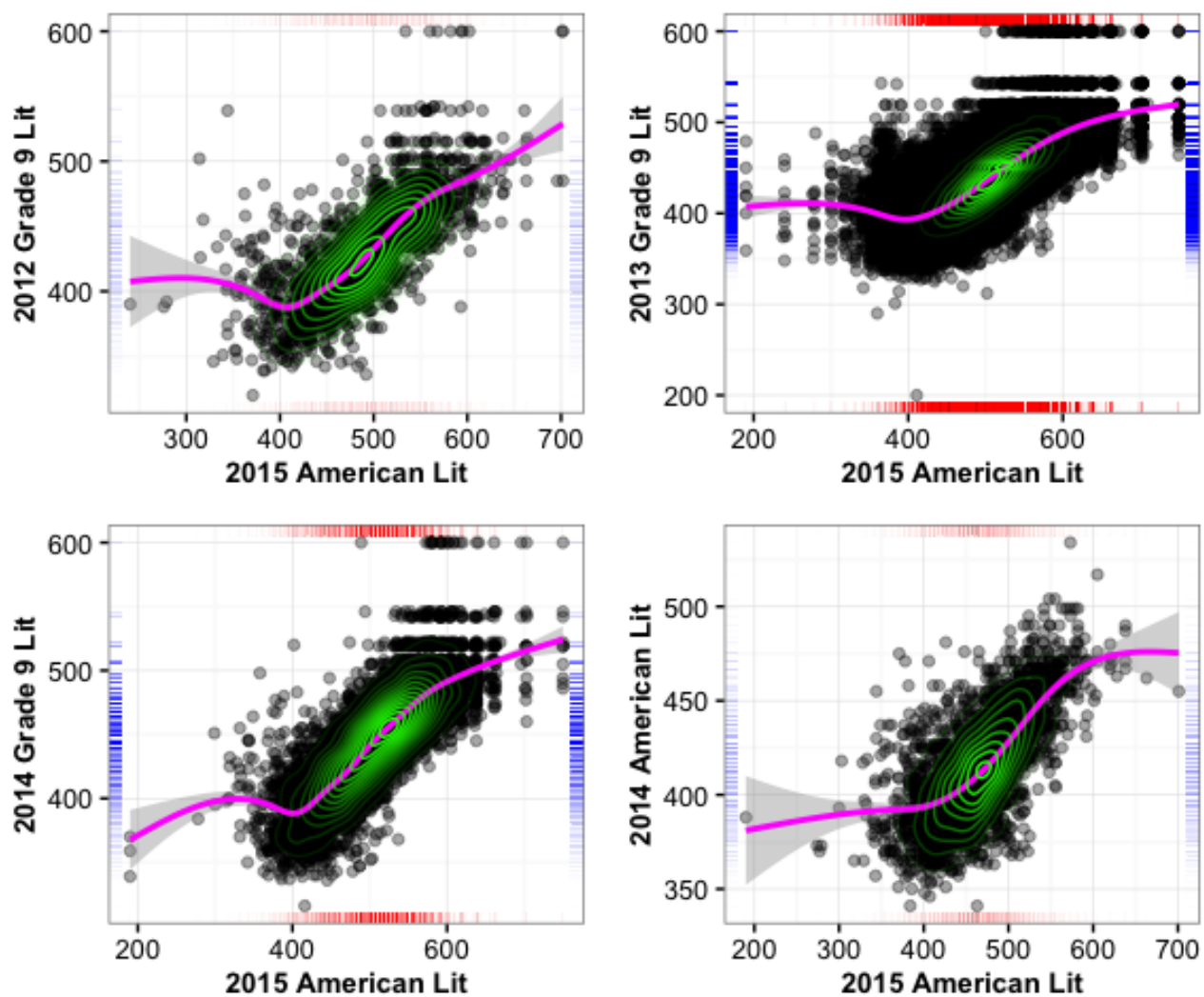


Fig. C.19: Conditional distribution(s) of current and prior scale scores: US History.

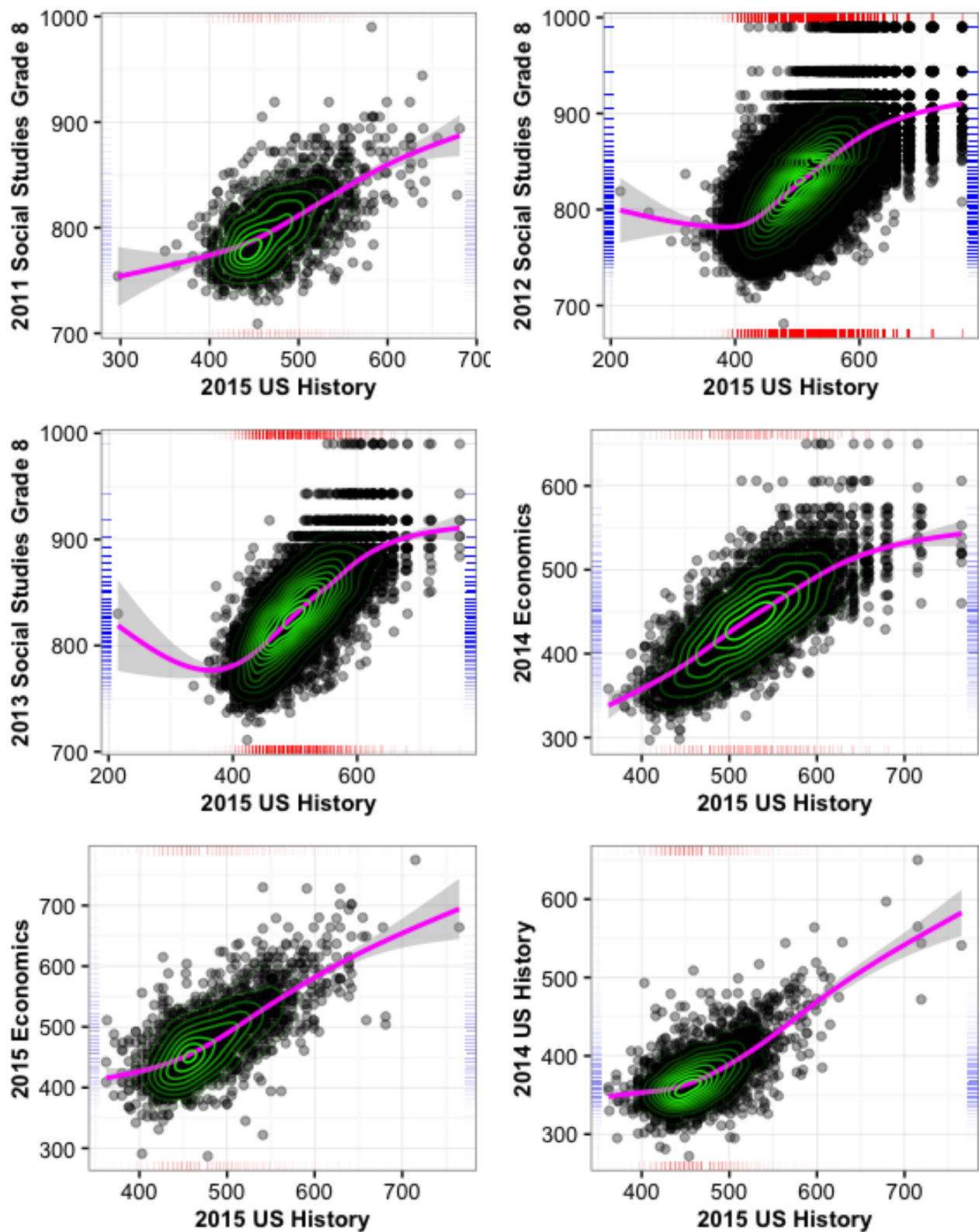


Fig. C.20: Conditional distribution(s) of current and prior scale scores: Economics.

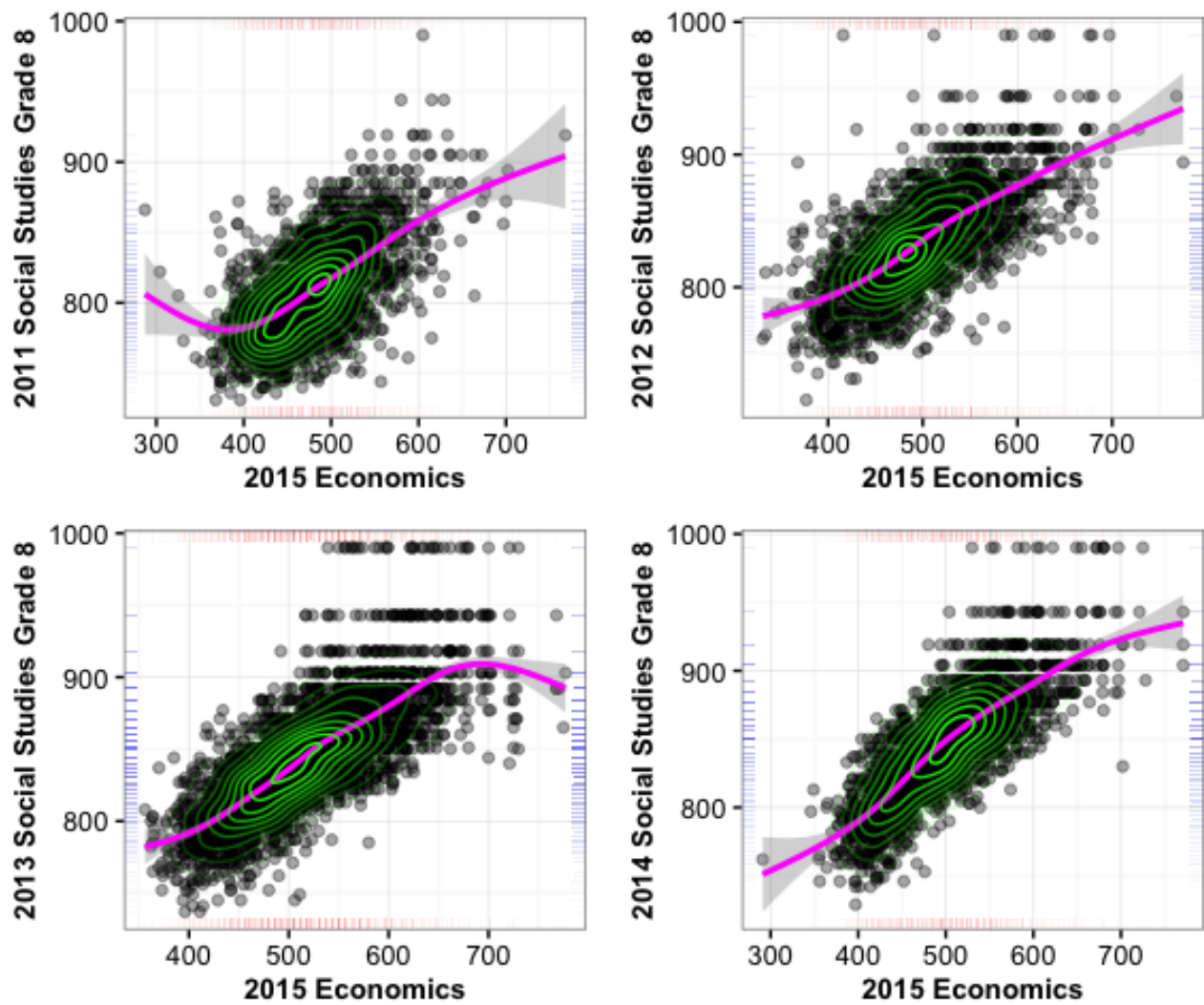


Fig. C.21: Conditional distribution(s) of current and prior scale scores: Economics *Continued*.

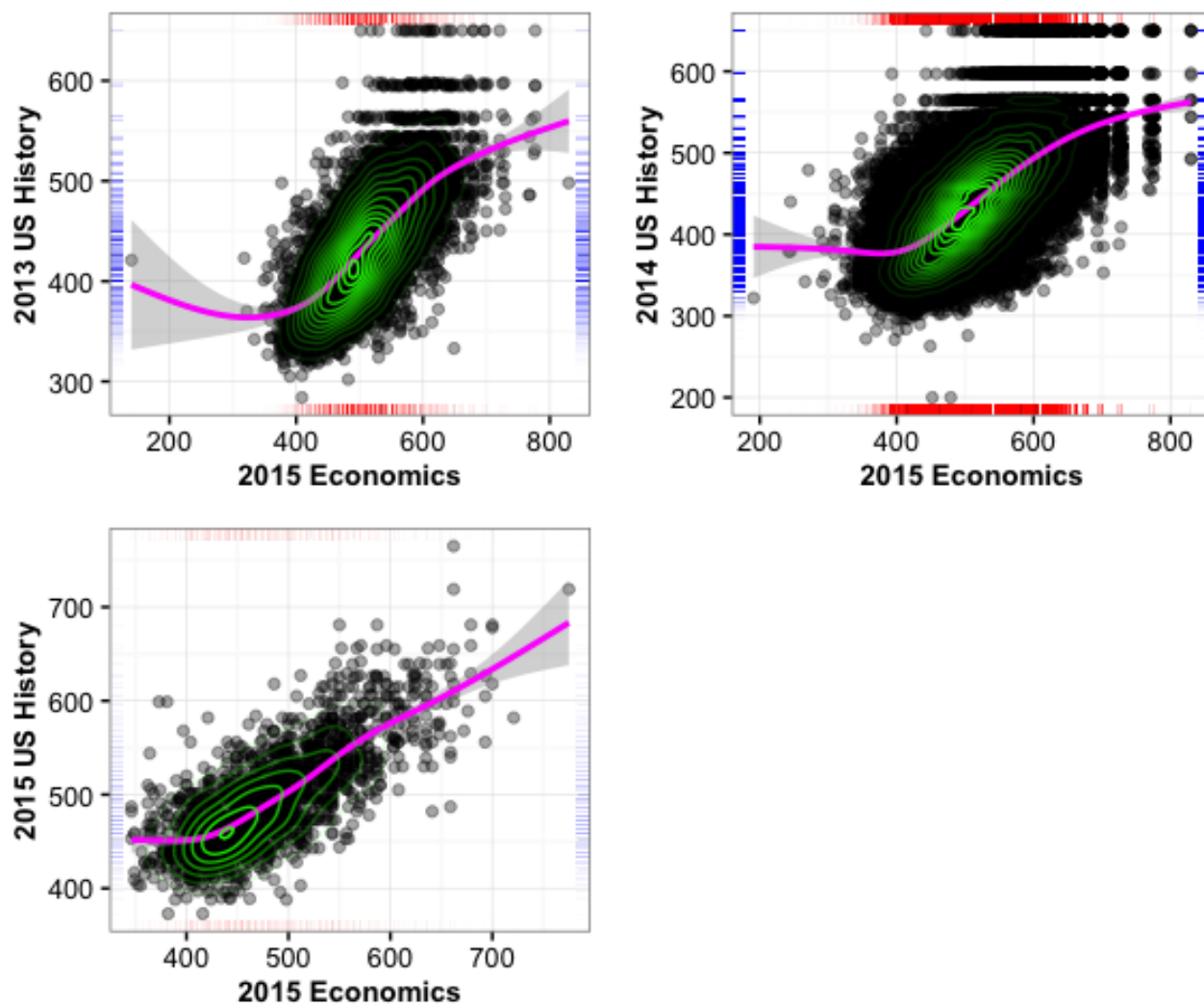


Fig. C.22: Conditional distribution(s) of current and prior scale scores: Biology.

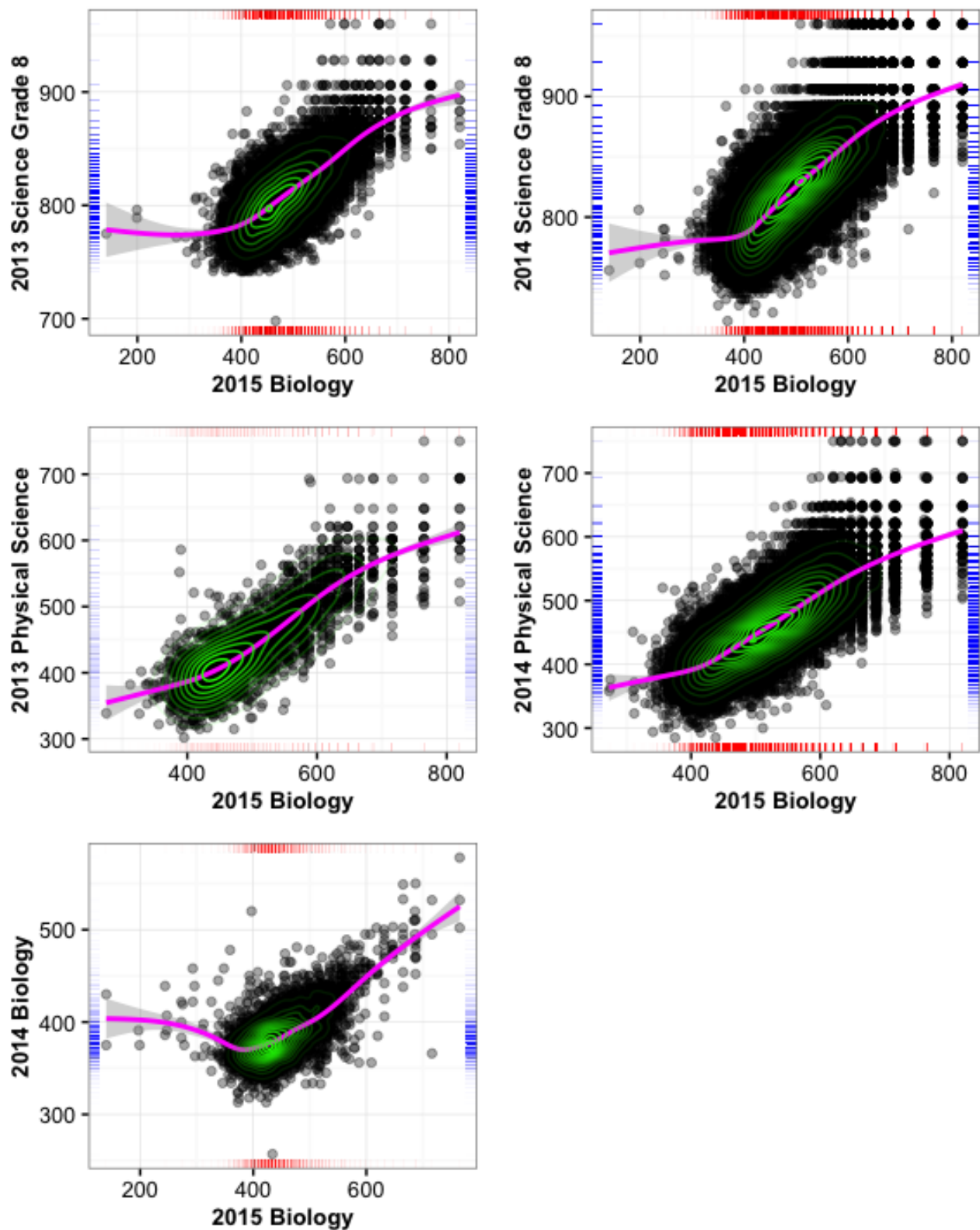


Fig. C.23: Conditional distribution(s) of current and prior scale scores: Physical Science.

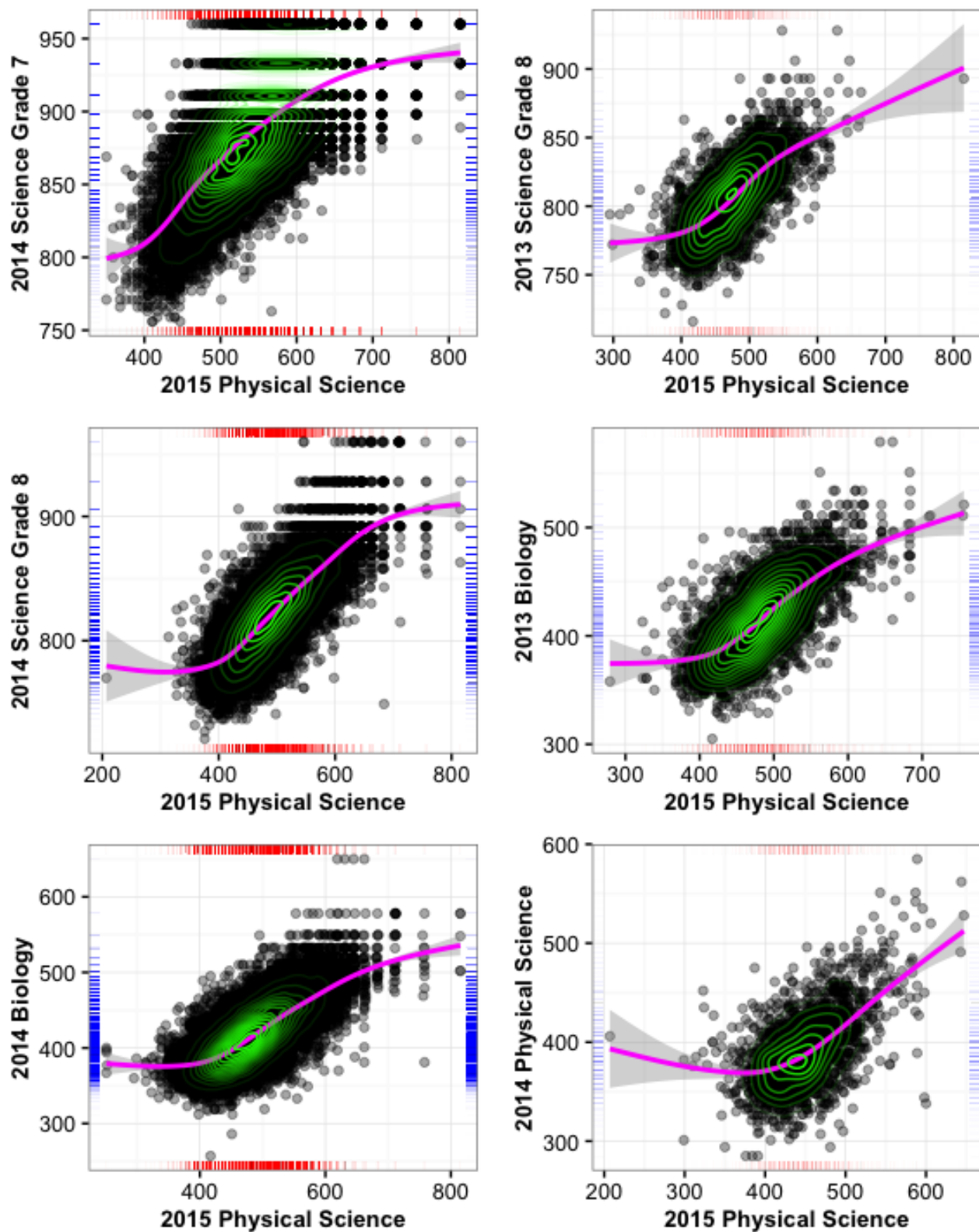


Fig. C.24: Conditional distribution(s) of current and prior scale scores: Coordinate Algebra.

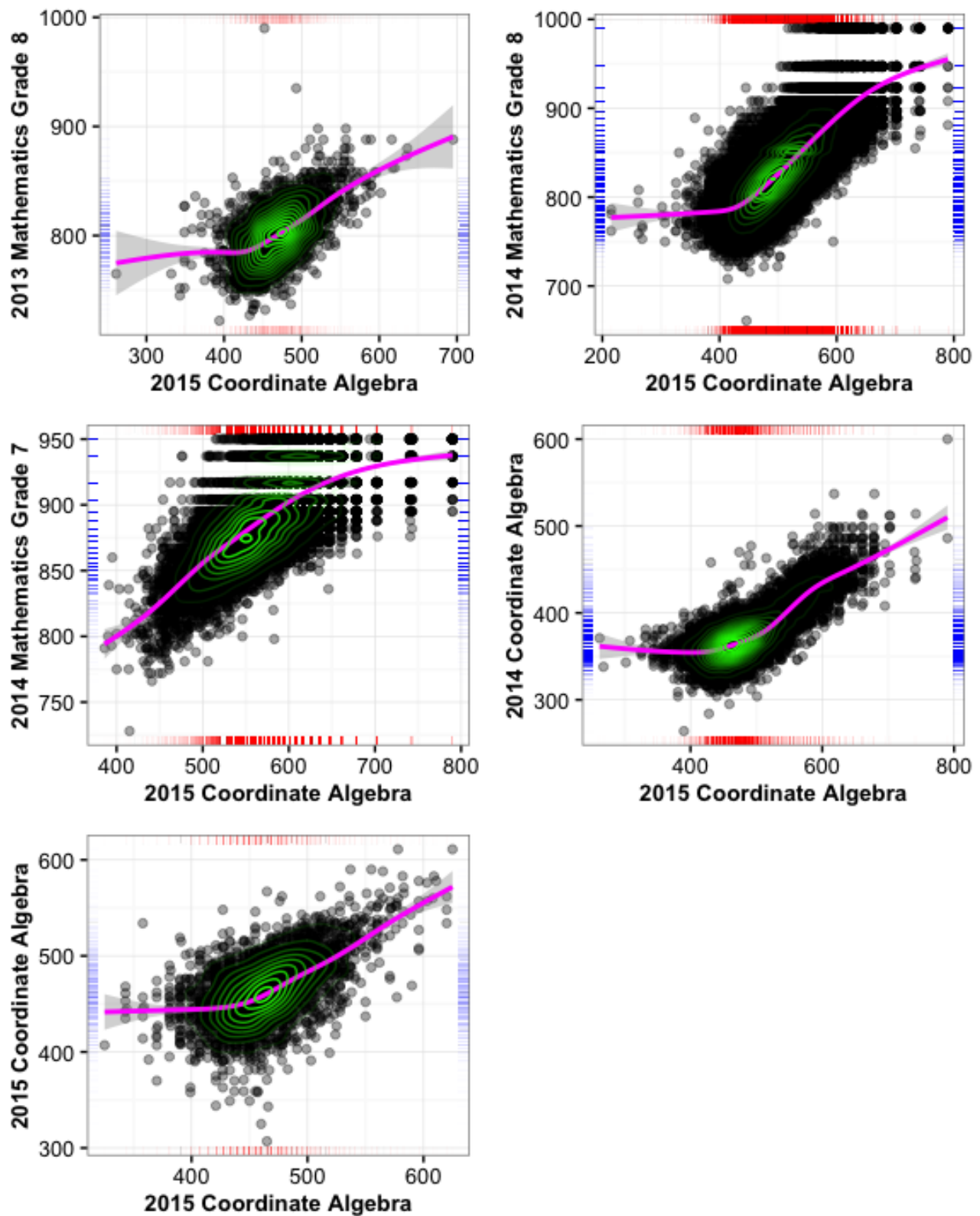
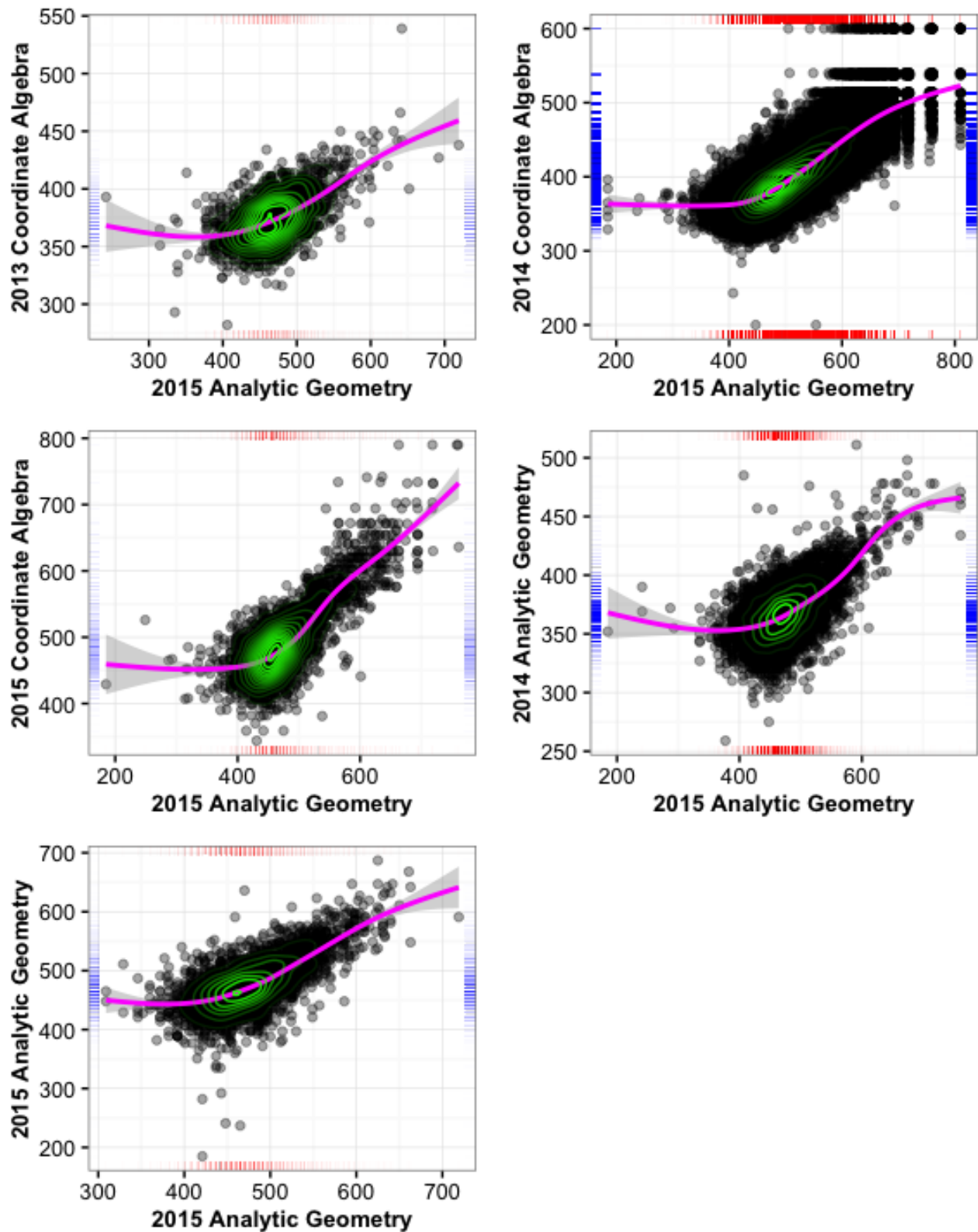


Fig. C.25: Conditional distribution(s) of current and prior scale scores: Analytic Geometry.



3 SGP Ranges for the Highest and Lowest Achieving Students

In order to isolate the impact of assessment ceilings/floors on student growth percentile (SGP) calculations, the following section provides box plots of the distribution of SGPs for the highest and lowest achieving students. We are specifically interested in the growth percentiles for students scoring at the highest/lowest obtainable scale score (HOSS/LOSS - i.e. the artificial test ceiling/floor) on the current year test. However, in order to assure that an adequate number of students are included, the first set of plots in each subsection uses, at a *minimum*, the highest/lowest 50 scores. Note that this roughly corresponds to the number of students used in the SGP model goodness of fit plots, and this is why these plots are provided as a starting point for this part of the investigation. All students with a score in these students' range of scores are included. Consequently, the number of students in each box plot may be greater than 50 (the exact number is shown at the margins in red text).

The second set of box plots in the following subsections isolate *only* those students scoring the HOSS/LOSS. These plots may then incorporate a widely varying number of students depending on the prevalence of a ceiling/floor in the current year. Note that because Georgia uses the SIMEX measurement error corrected SGP as the official SGP, that is the version used here.

The box plots provide several descriptive statistics. The dark line within the box marks the *median* SGP, while the ends ("hinges") of the boxes correspond to the first and third quartiles (the 25th and 75th percentiles). The upper whisker extends from the hinge to the highest value that is within $1.5 \times \text{IQR}$ of the hinge, where IQR is the inter-quartile range, or distance between the first and third quartiles. The lower whisker extends from the hinge to the lowest value within $1.5 \times \text{IQR}$ of the hinge. Data beyond the end of the whiskers are outliers and plotted as points. Evidence of a *lack* of either a ceiling or floor effect would be to have all high achieving students with SGPs near 99 and all low achieving students with SGPs near 1. That is, the desired visual evidence is a solid line at $\text{SGP} = 99/1$.

In the 2015 Georgia EOG SGP analyses, we see evidence of a very minor ceiling effects in some grades and content areas. However, these issues are present when considering the top 50 students' scores, but disappear when considering only the HOSS/LOSS scores and we see that all SGPs are above 90 (Figure C.27).

3.1 EOG Content Areas

Fig. C.26: SGP distributions for highest and lowest 50+ scale scores by content area and grade level.

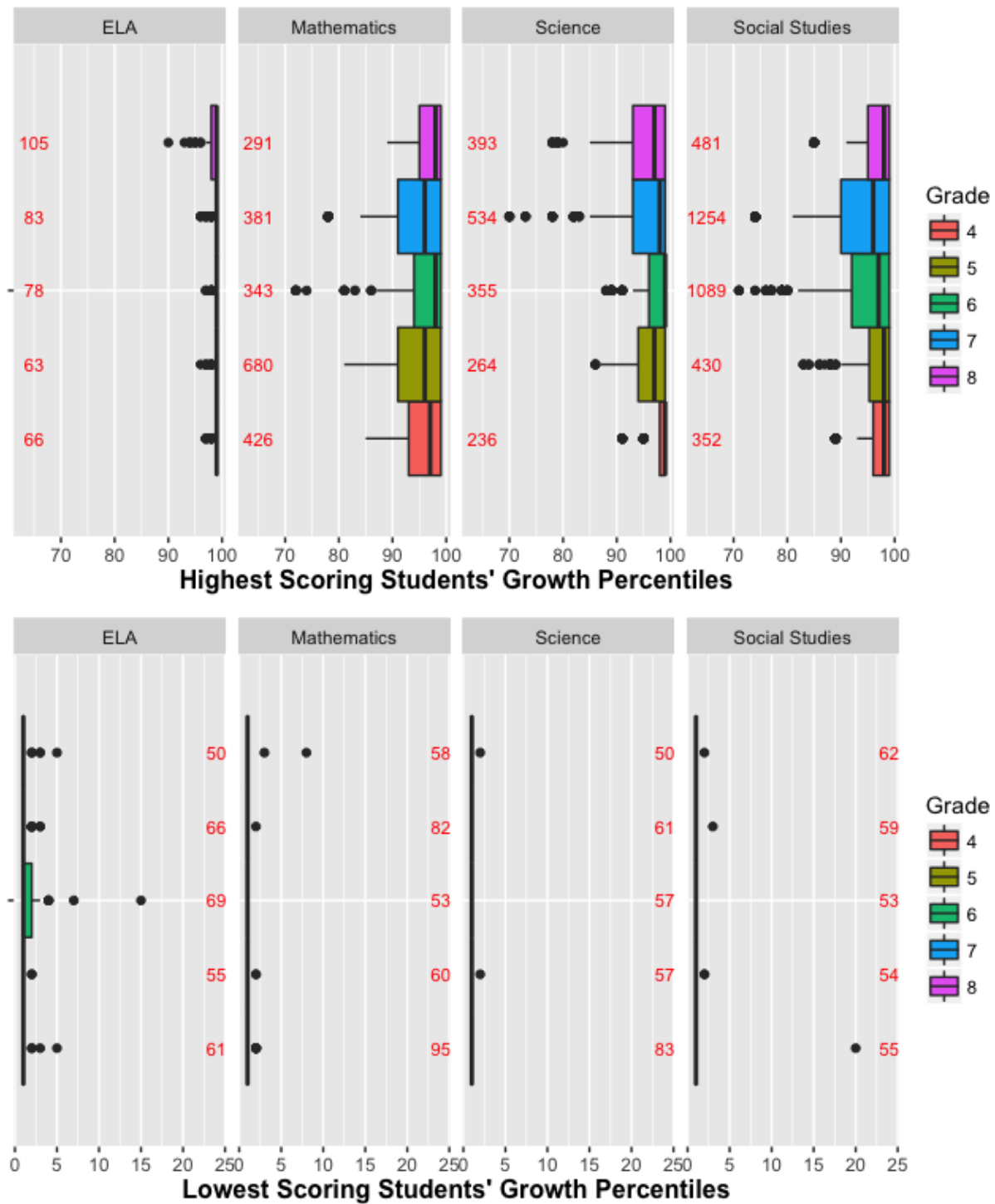
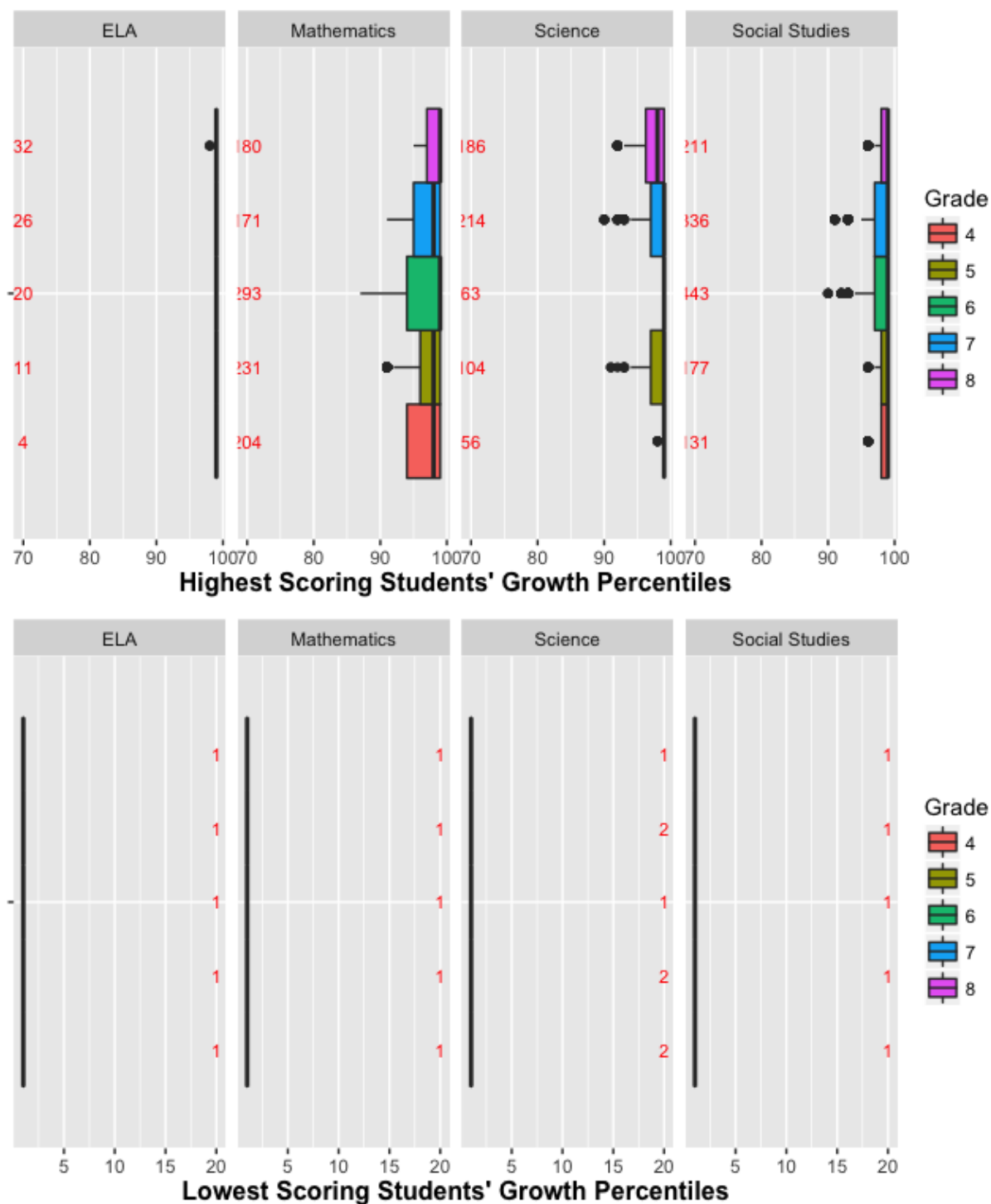


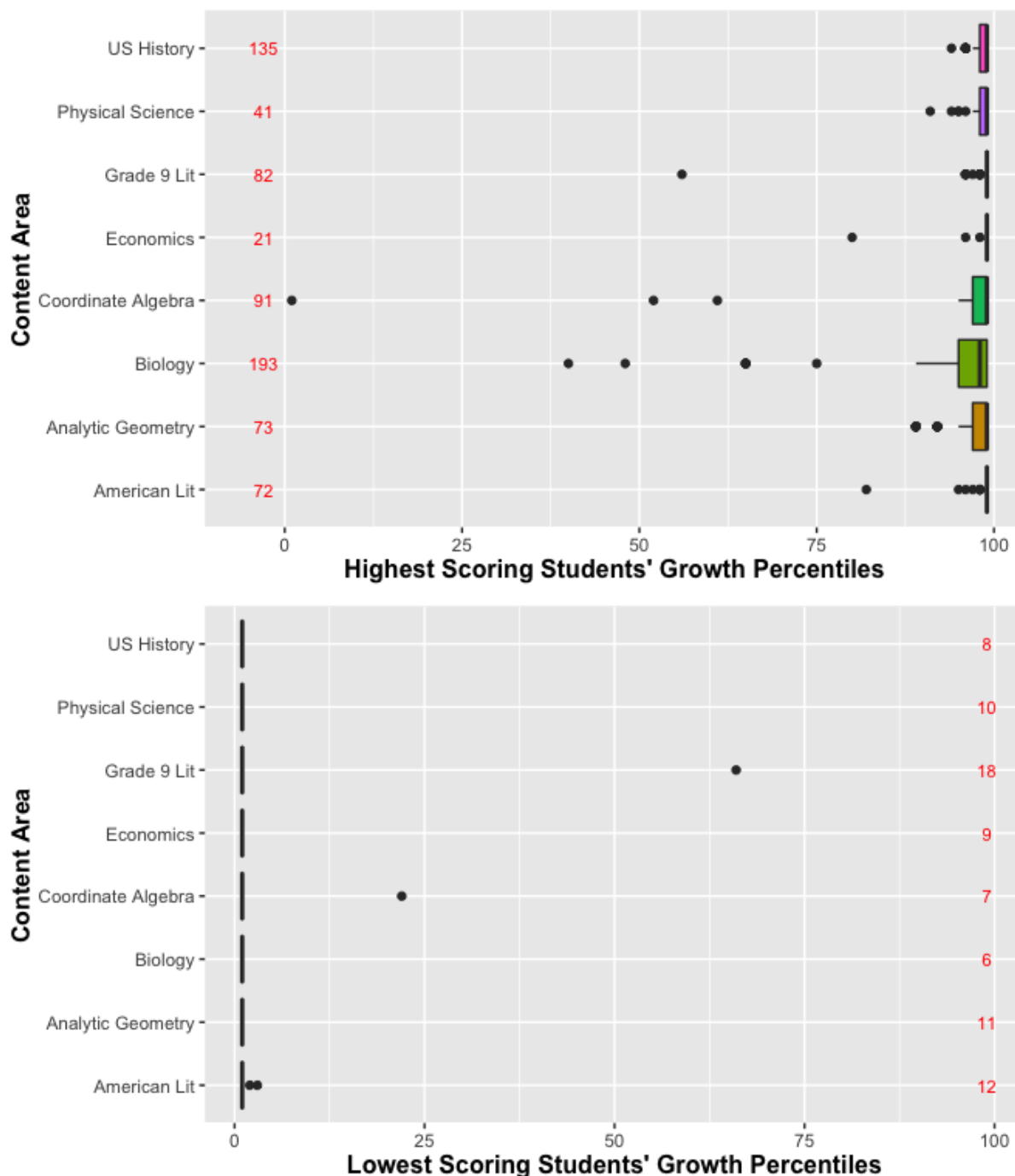
Fig. C.27: SGP distributions for the HOSS and LOSS scores by content area and grade level.



3.2 EOC Subjects

The end-of-course subject results are shown here *only for students scoring exactly the LOSS and HOSS*. There are several subjects for which potential ceiling effects are evident. These subjects are disaggregated further in order to more appropriately address these concerns.

Fig. C.28: EOC SGP distributions for the HOSS and LOSS scores by content area.



The end-of-course box plots can be disaggregated further by the most recent prior (i.e. their constituent norm groups). The following box plots disaggregate each EOC subject by norm groups. A pattern emerges in these plots that suggests that the problematic course progressions are those for course repeaters and those that included skipped years.

In the Grade 9 Literature plots below we see the potentially problematic single case from the course repeater analysis (highlighted earlier in the bottom-left panel of Figure C.17). This student has scored perfectly on both tests (which begs the question as to why this student was re-taking the test to begin with), and, given her relation to the rest of this cohort, the SGP model has estimated this to be a relatively common occurrence with an SGP of 56. This may be a fair assessment of the conditional probability of staying perfect, but is hard to justify as a description of this student's growth.

Fig. C.29: EOC SGP distributions for the HOSS and LOSS scores by norm group: 9th Grade Lit.

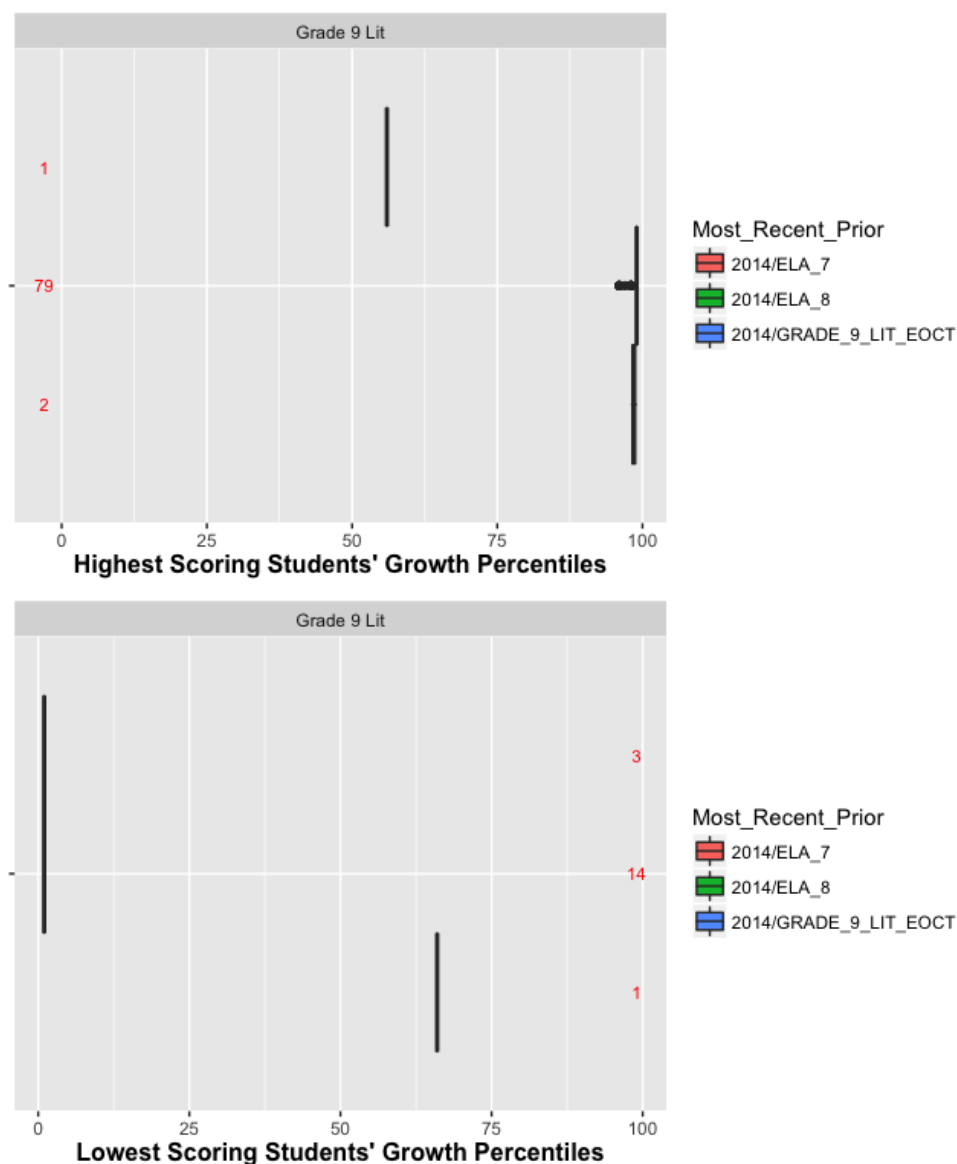


Fig. C.30: EOC SGP distributions for the HOSS and LOSS scores by norm group: American Lit.

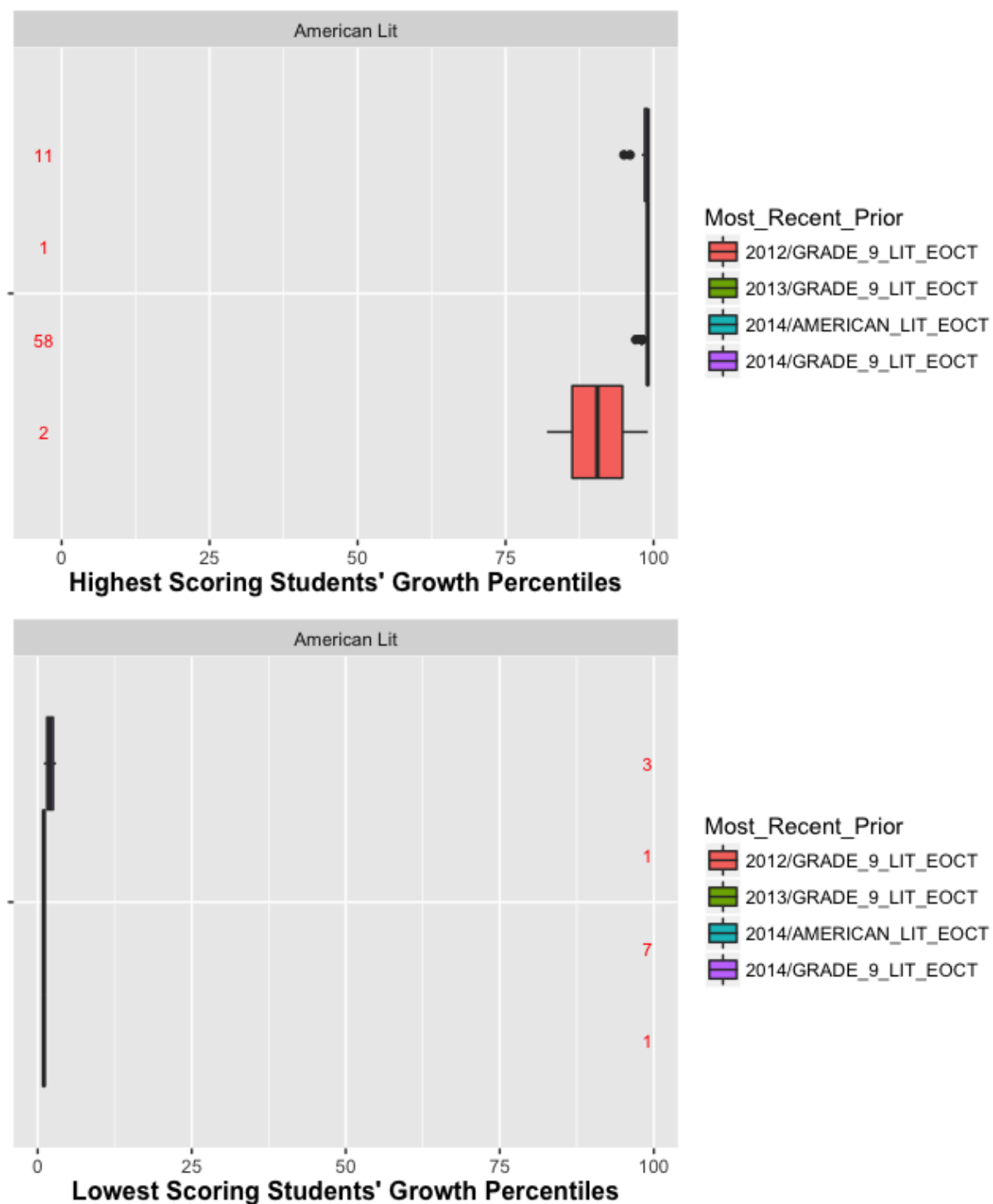


Fig. C.31: EOC SGP distributions for the HOSS and LOSS scores by norm group: US History.

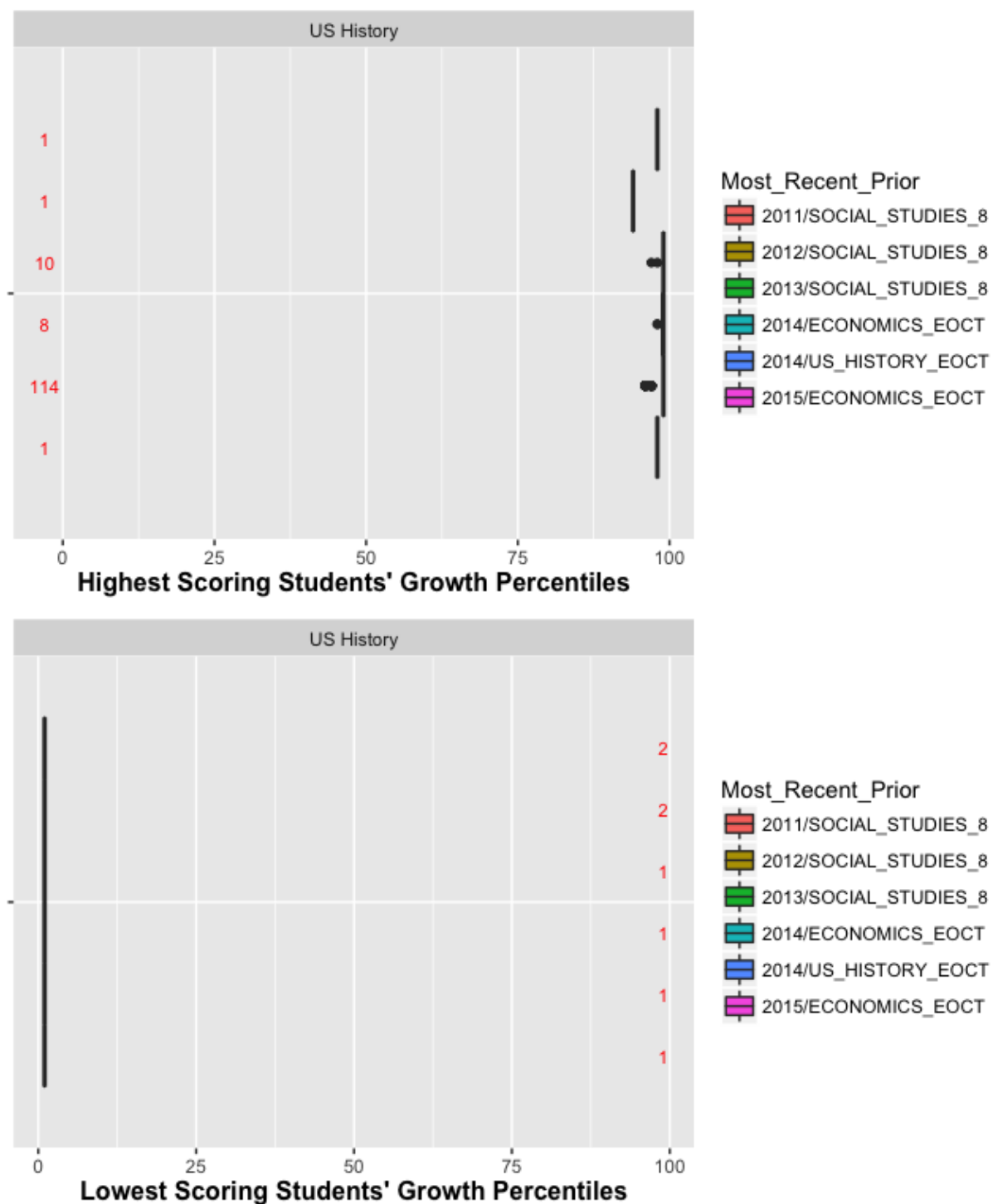
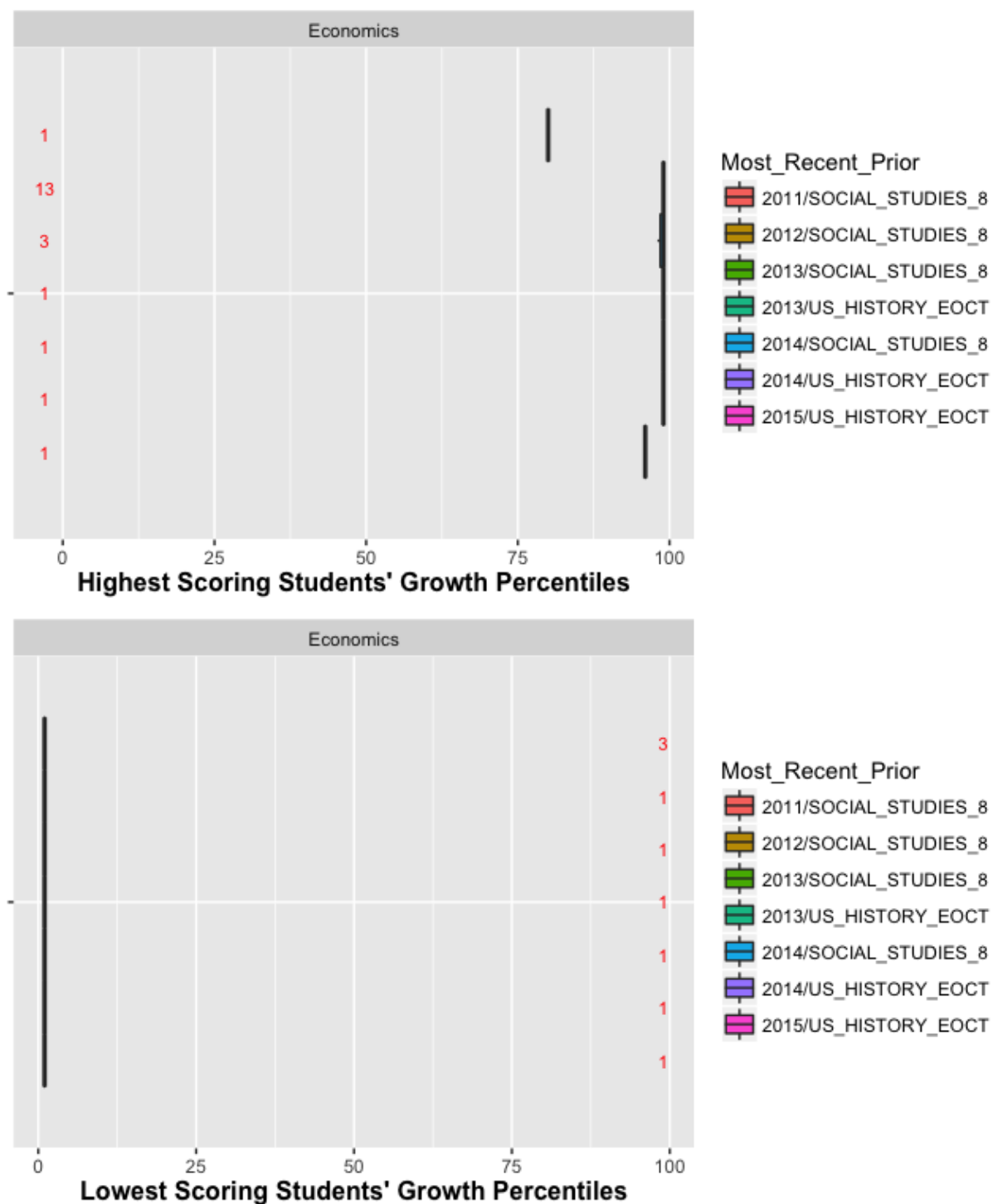


Fig. C.32: EOC SGP distributions for the HOSS and LOSS scores by norm group: Economics.



The plots for Biology are notable, particularly for the course repeater progression. This progression can also be seen in last plot of Figure C.22 above. Here we have isolated three students who scored the Milestones HOSS, 765. These student's prior scale scores are all relatively high compared to the other repeaters, yet there is a range in their three prior scores from 502 to 578. This translates to a range in SGPs from 97 to 40, and may be considered a minor ceiling effect for the highest attaining student.

Fig. C.33: EOC SGP distributions for the HOSS and LOSS scores by norm group: Biology.

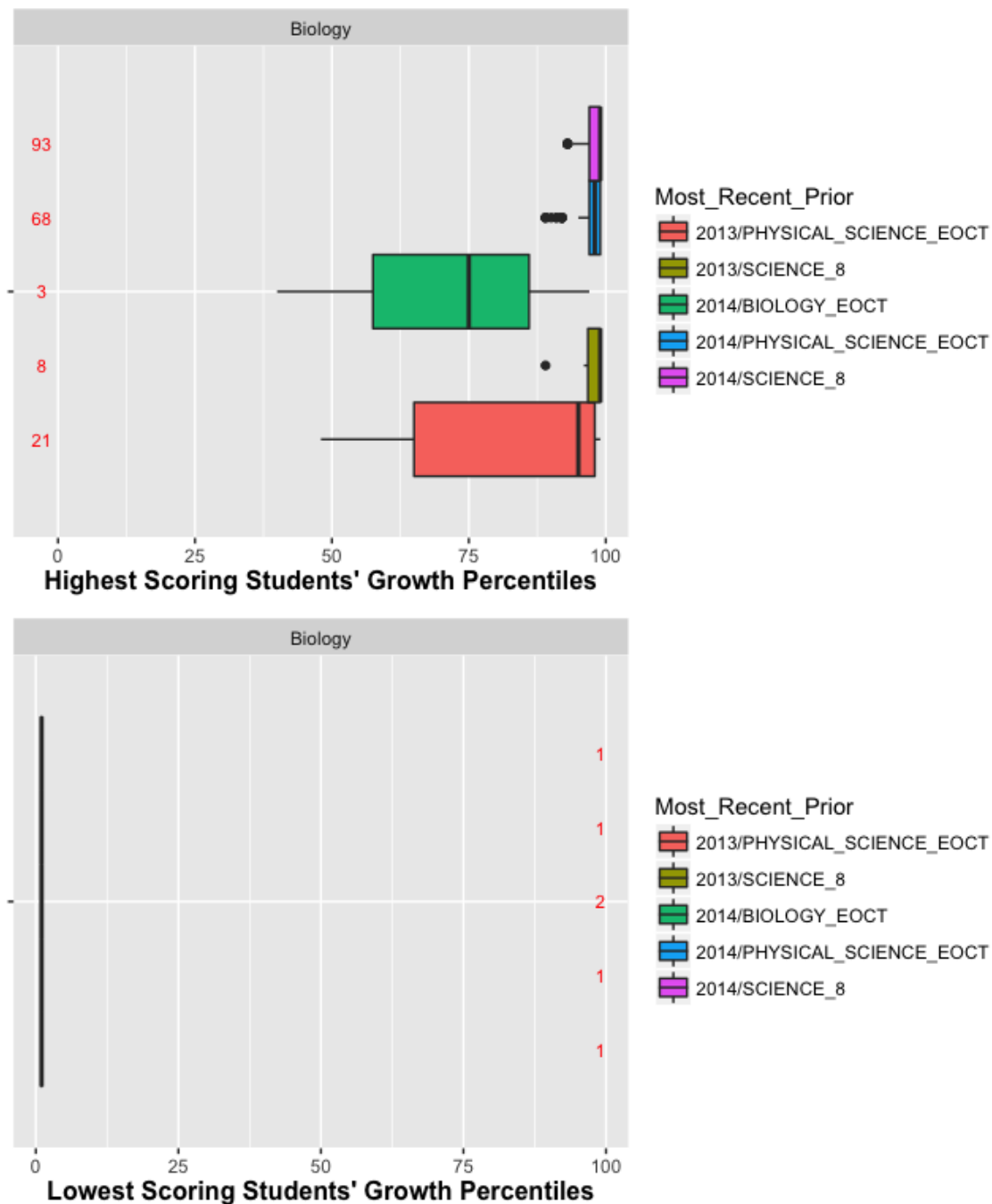
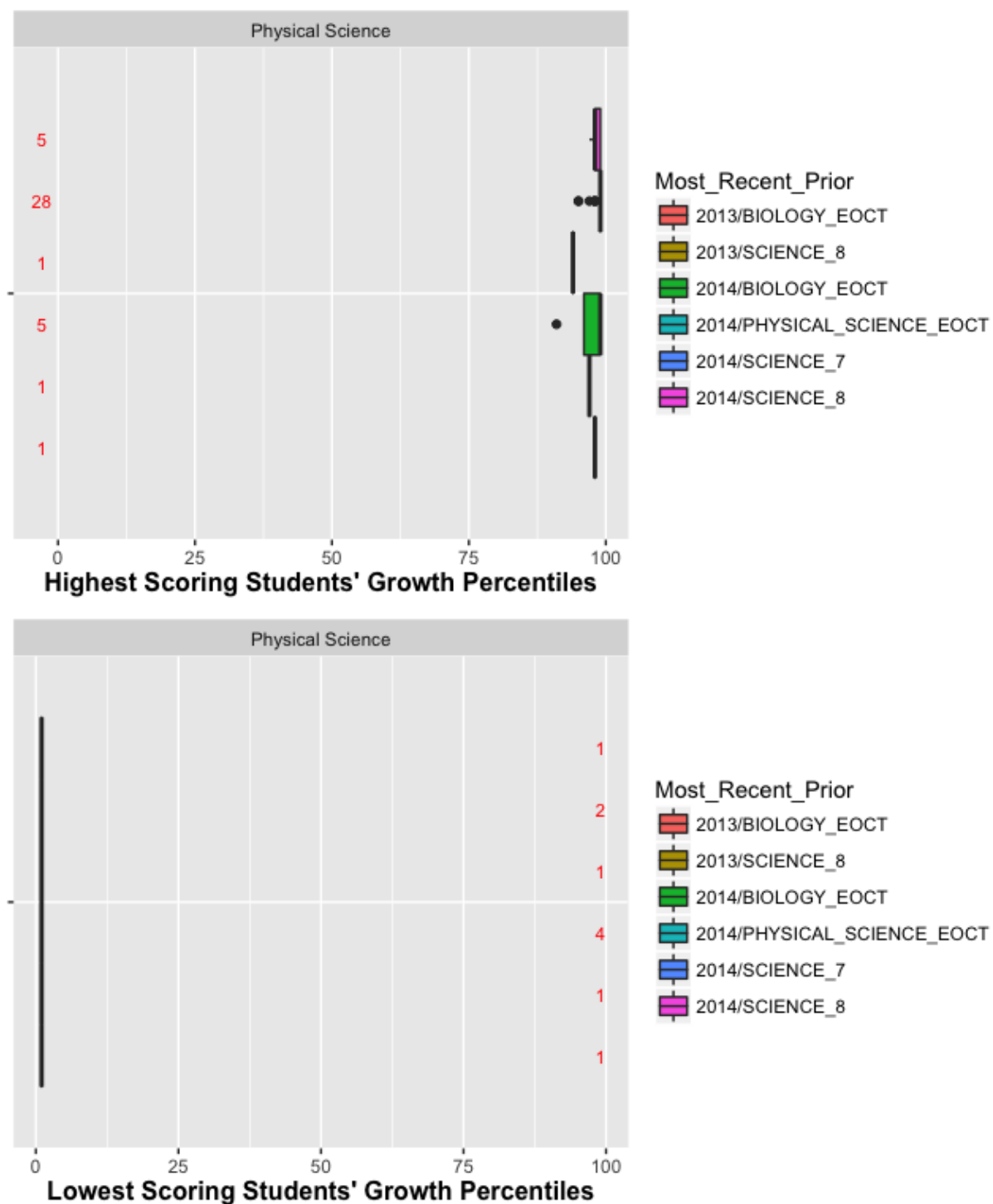


Fig. C.34: EOC SGP distributions for the HOSS and LOSS scores by norm group: Physical Science.



The plots for Coordinate Algebra are notable, particularly for the 2014 course repeater progression. This progression can also be seen in second-to-last plot of Figure C.24 above. Here we have isolated two students who scored the HOSS, 790. One student's prior scale score was relatively low, although still quite high for the group overall at 480. This student's SGP was estimated to be 99. The other student had a prior score of 600, which was perfect on the 2014 test (which begs the question of why this student was repeating the course to begin with!). This student's SGP was estimated to be 1, which is probably an inaccurate assessment of this students growth since they have attained perfect scores two years in a row.

Fig. C.35: EOC SGP distributions for the HOSS and LOSS scores by norm group: Coordinate Algebra.

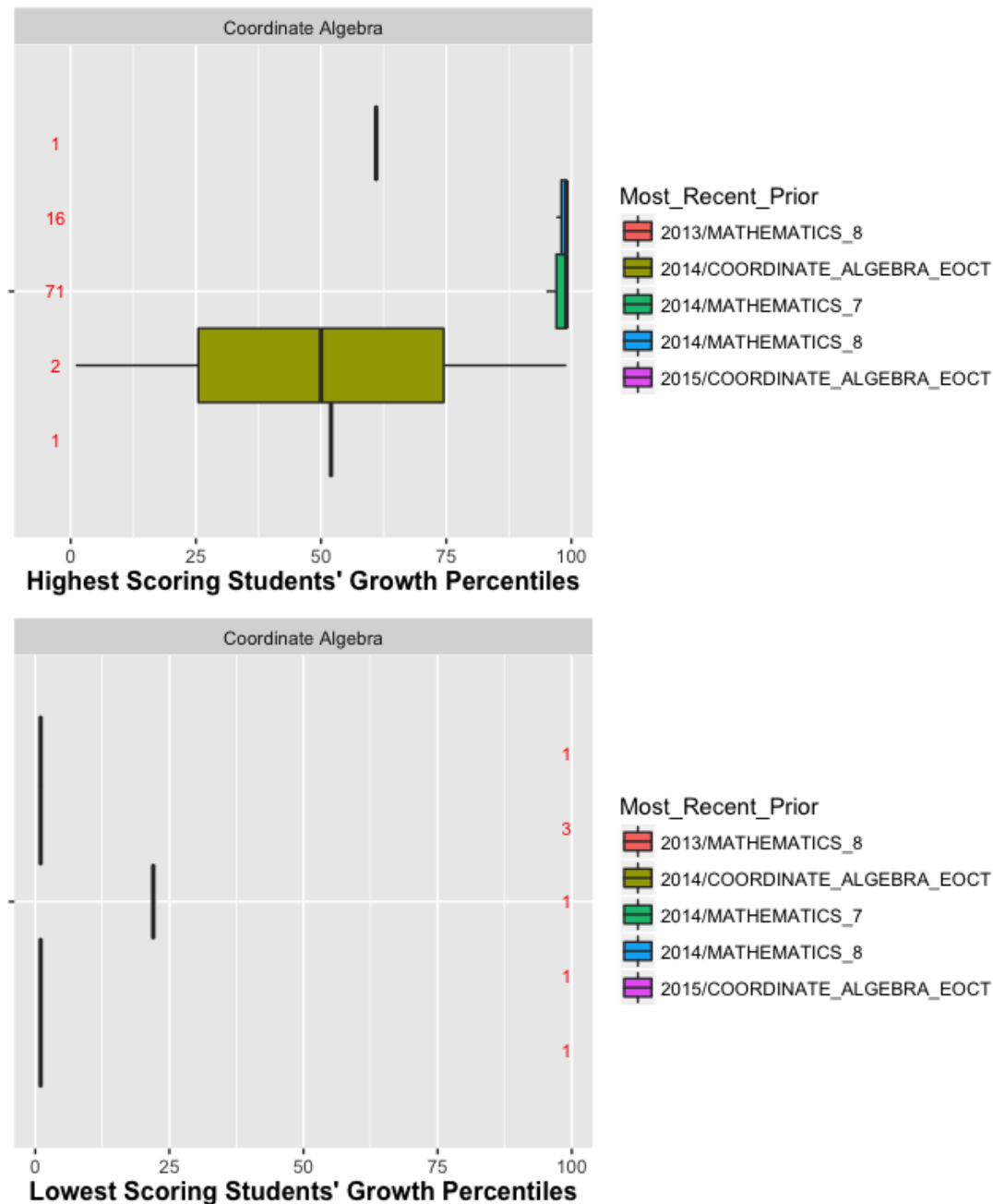
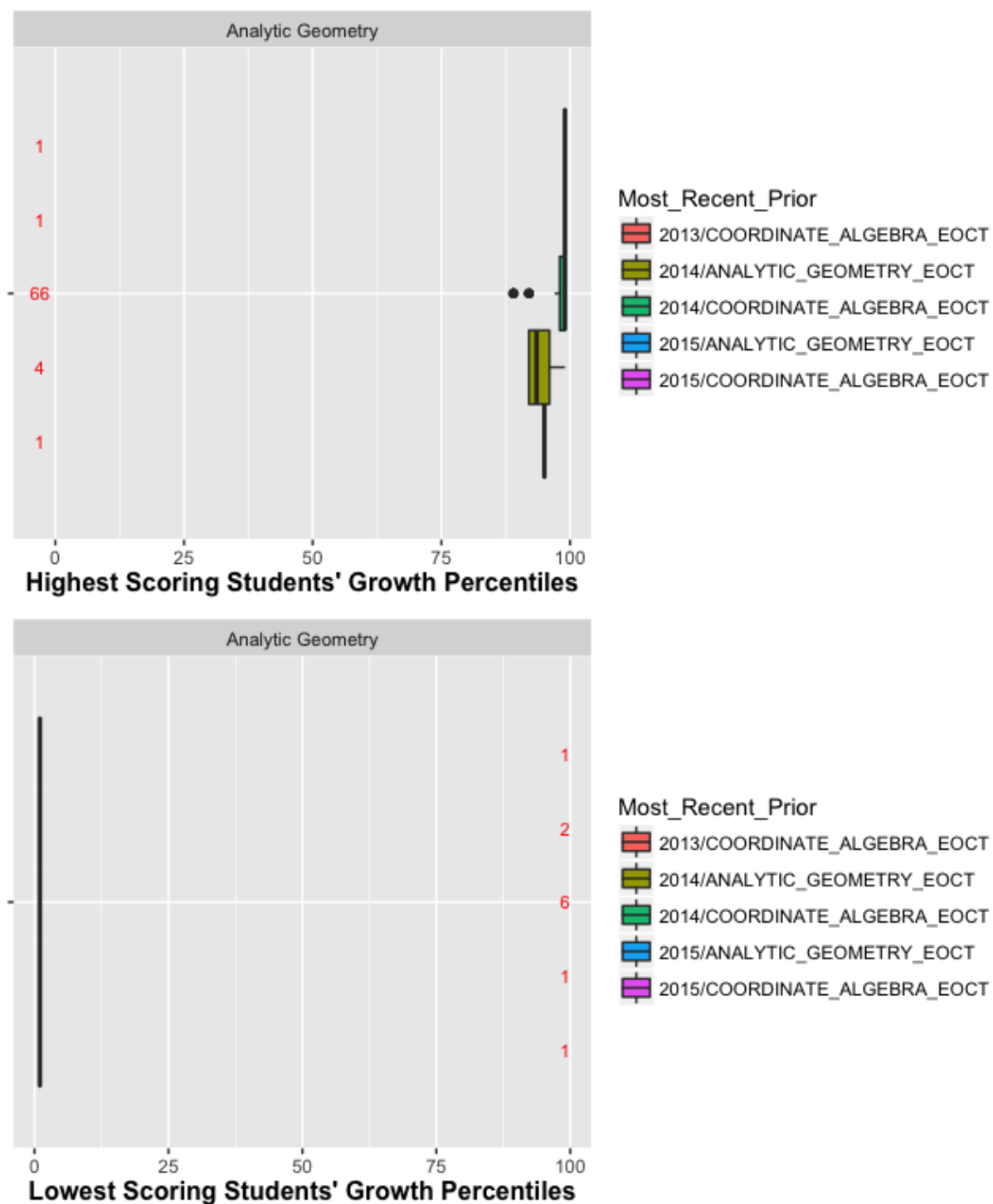


Fig. C.36: EOC SGP distributions for the HOSS and LOSS scores by norm group: Analytic Geometry.



4 Discussion

Overall there is little evidence of floor or ceiling effects in the 2015 Georgia Milestones SGP analyses. Some scores for EOC test repeaters and skip year progressions suggest that a small number of minor problems may exist that could require changes.

When ceiling or floor effects are encountered, there are several ways in which they can be “corrected” manually or analytically. These include (but not limited to):

1. Convert all students scoring at the HOSS (LOSS) to 99 (1).
2. Run SGP analyses with more granular scores. For example, many tests that use Item Response Theory (IRT) to analyse test results provide scaled scores that enforce an artificial ceiling (floor), but also have more granular achievement scores available (IRT θ estimates).
3. Leave the results without a correction.